[Continuation of Memoir No. 146, "A Memoir on Curves of the Third Order." The present block (PDF pp. 401–432) carries the memoir from Article No. 3 through the recapitulation in Article No. 41.]

It will be presently seen that the analytical theory leads to the consideration of a line IJ (not represented in the figure); the line in question is the polar of F (or E) with respect to the conic which is the first or conic polar of E (or F) with respect to any syzygetic cubic. The line IJ is a tangent of the Pippian, and moreover the lines EF and IJ are conjugate polars of a curve of the third class syzygetically connected with the Pippian and Quippian, and which is moreover such that its Hessian is the Pippian.

Article Nos. 3 to 19. Analytical investigations, comprising the proof of the theorems, Article No. 2.

3. The analytical theory possesses considerable interest. Take as the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

then the equation of the Hessian is

$$HU = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0;$$

and the equation of the Pippian in line coordinates (that is, the equation which expresses that $\xi x+\eta y+\zeta z = 0$ is a tangent of the curve) is

$$PU = -l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\,\xi\eta\zeta = 0.$$

The equation of the Quippian in line coordinates is

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(5 + 4l^3)\,\xi\eta\zeta = 0;$$

and the values of the two invariants of the cubic form are

$$S = -l + l^4,$$

$$T = 1 - 20l^3 - 8l^6,$$

values which give identically,

$$T^2 - 64S^3 = (1 + 8l^3)^4,$$

the last-mentioned function being in fact the discriminant.

4. Suppose now that (X, Y, Z) are the coordinates of the point E, and (X', Y', Z') the coordinates of the point F; then the equations which express that these points are conjugate poles of the cubic, are

$$\begin{aligned} XX' + l(YZ' + Y'Z) &= 0,\\ YY' + l(ZX' + Z'X) &= 0,\\ ZZ' + l(XY' + X'Y) &= 0; \end{aligned}$$

and by eliminating from these equations, first (X', Y', Z'), and then (X, Y, Z), we find

$$\begin{aligned} l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)\,XYZ &= 0,\\ l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)\,X'Y'Z' &= 0, \end{aligned}$$

which shows that the points E, F are each of them points of the Hessian.

5. I may notice, in passing, that the preceding equations give rise to a somewhat singular *unsymmetrical* quadratic transformation of a cubic form. In fact, the second and third equations give

$$X' : Y' : Z' = YZ - l^2X^2 : l^2XY - lZ^2 : l^2ZX - lY^2.$$

And substituting these values for X', Y', Z' in the form

$$l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)X'Y'Z',$$

the result must contain as a factor

$$l^2(X^3+Y^3+Z^3)-(1+2l^3)XYZ;$$

the other factor is easily found to be

$$-l^3(l^3(X^3+Y^3+Z^3)+3lXYZ).$$

Several of the formulae given in the sequel conduct in like manner to unsymmetrical transformations of a cubic form.

6. I remark also, that the last-mentioned system of equations gives, symmetrically,

$$X'^2 : Y'^2 : Z'^2 : Y'Z' : Z'X' : X'Y'$$

$$= YZ-l^2X^2 \;:\; ZX-l^2Y^2 \;:\; XY-l^2Z^2 \;:\; l^2YZ-lX^2 \;:\; l^2ZX-lY^2 \;:\; l^2XY-lZ^2;$$

and it is, I think, worth showing how, by means of these relations, we pass from the equation between X', Y', Z' to that between X, Y, Z. In fact, representing, for shortness, the foregoing relations by

$$X'^2 : Y'^2 : Z'^2 : Y'Z' : Z'X' : X'Y' = A : B : C : F : G : H,$$

we may write

$$X'=AF=GH, \quad Y'=BG=HF, \quad Z'=CH=FG, \quad ABC=FGH;$$

and thence

$$X'^2=AF\cdot GH, \quad Y'^2=BG\cdot HF, \quad Z'^2=CH\cdot FG, \quad X'Y'Z'=FGH;$$

hence

$$l^2(X'^3+Y'^3+Z'^3)-(1+2l^3)X'Y'Z' = FGH\{l^2(AGH+BHF+CFG)-(1+2l^3)FGH\}.$$

But we have

$$\begin{aligned} l^2(AGH+BHF+CFG) &= -(2l^3+l^6)(X^3+Y^3+Z^3)XYZ+(l^3+2l^6)(Y^3Z^3+Z^3X^3+X^3Y^3),\\ -(1+2l^3)FGH &= (l^3+2l^6)(X^3+Y^3+Z^3)XYZ+(l^4+2l^7)(Y^3Z^3+Z^3X^3+X^3Y^3)\\ &\quad + l^3(1-l^3)(1+2l^3)X'Y'Z'; \end{aligned}$$

and thence

$$l^2(AGH+BHF+CFG)-(1+2l^3)FGH = -l^3(1-l^3)\{l^2(X^3+Y^3+Z^3)XYZ-(1+2l^3)X^2Y^2Z^2\};$$

and finally,

$$l^2(X'^3+Y'^3+Z'^3)-(1+2l^3)X'Y'Z' = l^3(-1+l^3)(lXZ-Y^2)(lZY-X^2)(lXY-Z^2)\times\{l^2(X^3+Y^3+Z^3)-(1+2l^3)XYZ\}.$$

We have also, identically,

$$ABC-FGH=\tfrac{1}{l}(-l+l^4)XYZ\{l^2(X^3+Y^3+Z^3)-(1+2l^3)XYZ\},$$

which agrees with the relation $\dfrac{FGH}{ABC}=0$.

7. Before going further, it will be convenient to investigate certain relations which exist between the quantities (X,Y,Z), (X',Y',Z'), connected as before by the equations

$$\begin{aligned} XX'+l(YZ'+Y'Z)&=0,\\ YY'+l(ZX'+Z'X)&=0,\\ ZZ'+l(XY'+X'Y)&=0, \end{aligned}$$

and the quantities

$$\begin{aligned} \xi &= YZ' - Y'Z, & \alpha &= XX' = -\tfrac{1}{l}(YZ' + Y'Z), \\ \eta &= ZX' - Z'X, & \beta &= YY' = -\tfrac{1}{l}(ZX' + Z'X), \\ \zeta &= XY' - X'Y, & \gamma &= ZZ' = -\tfrac{1}{l}(XY' + X'Y). \end{aligned}$$

We have identically,

$$2XX'(YZ' - Y'Z) + (XY' + X'Y)(ZX' - Z'X) + (ZX' + Z'X)(XY' - X'Y) = 0;$$

or expressing in terms of $\xi, \eta, \zeta, \alpha, \beta, \gamma$ the quantities which enter into this equation, and forming the analogous equations, we have

$$\begin{aligned} 2l\alpha\xi - \gamma\eta - \beta\zeta &= 0, \\ -\gamma\xi + 2l\beta\eta - \alpha\zeta &= 0, \\ -\beta\xi - \alpha\eta + 2l\gamma\zeta &= 0. \end{aligned} \tag{A}$$

We have also

$$X^2Y'Z' - X'^2YZ = \tfrac{1}{2}\{-(XY' + X'Y)(ZX' - Z'X) + (ZX' + Z'X)(XY' - X'Y)\},$$

and thence in like manner,

$$\begin{aligned} X^2Y'Z' - X'^2YZ &= \tfrac{1}{2l}(\gamma\eta - \beta\zeta), \\ Y^2Z'X' - Y'^2ZX &= \tfrac{1}{2l}(\alpha\zeta - \gamma\xi), \\ Z^2X'Y' - Z'^2XY &= \tfrac{1}{2l}(\beta\xi - \alpha\eta). \end{aligned} \tag{B}$$

Again, we have

$$\begin{aligned} (YZ' - Y'Z)^2 &= (YZ' + Y'Z)^2 - 4YY'ZZ', \\ (ZX' - Z'X)(XY' - X'Y) &= -(ZX' + Z'X)(XY' + X'Y) + 2XX'(YZ' + Y'Z); \end{aligned}$$

and thence

$$\begin{aligned} \xi^2 &= l^2\alpha^2 - 4\beta\gamma, \\ \eta\zeta &= -l^2\alpha^2 - \tfrac{2}{l}\alpha, \qquad \text{(C)} \\ \xi\eta &= -l^2\gamma^2 - \tfrac{2}{l}\gamma, \\ \xi\zeta &= -l^2\beta^2 - \tfrac{2}{l}\beta; \qquad \text{(1)} \end{aligned}$$

and conversely

$$\begin{aligned} \tfrac{1}{l}(1 + 8l^3)\alpha^2 &= \xi^2 - 4l^2\eta\zeta, \\ \tfrac{1}{l}(1 + 8l^3)\beta^2 &= \eta^2 - 4l^2\zeta\xi, \qquad \text{(D)} \\ \tfrac{1}{l}(1 + 8l^3)\gamma^2 &= \zeta^2 - 4l^2\xi\eta, \\ -\tfrac{1}{l}(1 + 8l^3)\,\beta\gamma &= 2l\xi^2 + \eta\zeta, \qquad \text{(2)} \\ -\tfrac{1}{l}(1 + 8l^3)\,\gamma\alpha &= 2l\eta^2 + \zeta\xi, \qquad \text{(3)} \\ -\tfrac{1}{l}(1 + 8l^3)\,\alpha\beta &= 2l\zeta^2 + \xi\eta. \qquad \text{(4)} \end{aligned}$$

8. It is obvious that

$$\xi x + \eta y + \zeta z = 0$$

is the equation of the line EF joining the two conjugate poles, and it may be shown that

$$\alpha x + \beta y + \gamma z = 0$$

is the equation of the line IJ, which is the polar of E with respect to a conic which is the first or conic polar of F with respect to any syzygetic cubic. In fact, the equation of a syzygetic cubic will be $x^3+y^3+z^3+6\lambda xyz=0$, where λ is arbitrary, and the equation of the line in question is

$$(X\partial_x+Y\partial_y+Z\partial_z)(X'\partial_x+Y'\partial_y+Z'\partial_z)\big(x^3+y^3+z^3+6\lambda xyz\big)=0;$$

or developing,

$$XX'x+YY'y+ZZ'z+\lambda\big\{(YZ'+Y'Z)x+(ZX'+Z'X)y+(XY'+X'Y)z\big\}=0;$$

and the function on the left-hand side is

$$\Big(1-\tfrac{\lambda}{l}\Big)(\alpha x+\beta y+\gamma z),$$

which proves the theorem.

9. The equations (A) by the elimination of (ξ,η,ζ), give

$$-l(\alpha^3+\beta^3+\gamma^3)+(-1+4l^3)\,\alpha\beta\gamma=0,$$

which shows that the line IJ is a tangent of the Pippian: the proof of the theorem is given in this place because the relation just obtained between α,β,γ is required for the proof of some of the other theorems.

10. To find the coordinates of the point G in which the line EF joining two conjugate poles again meets the Hessian.

We may take for the coordinates of G,

$$uX+vX',\ uY+vY',\ uZ+vZ';$$

and, substituting in the equation of the Hessian, the terms containing u^2, v^2 disappear, and the ratio $u:v$ is determined by a simple equation. It thus appears that we may write

$$\begin{aligned}u&=-3l^2(XX'^2+YY'^2+ZZ'^2)+(1+2l^3)(Y'Z'X+Z'X'Y+X'Y'Z),\\ v&=3l^2(X^2X'+Y^2Y'+Z^2Z')-(1+2l^3)(YZX'+ZXY'+XYZ');\end{aligned}$$

hence introducing, as before, the quantities $\xi,\eta,\zeta,\alpha,\beta,\gamma$, we find

$$uX+vX'=3l^2(\gamma\eta-\beta\zeta)+(1+2l^3)(X^2Y'Z'-X'^2YZ);$$

but from the first of the equations (B),

$$X^2Y'Z'-X'^2YZ=\tfrac{1}{2l}(\gamma\eta-\beta\zeta),$$

and therefore the preceding value of $uX+vX'$ becomes

$$\Big(3l^2+\tfrac{1+2l^3}{2l}\Big)(\gamma\eta-\beta\zeta),$$

which is equal to

$$\frac{-1+4l^3}{2l}(\gamma\eta-\beta\zeta).$$

Hence throwing out the constant factor, we find, for the coordinates of the point G, the values

$$\gamma\eta-\beta\zeta,\quad \alpha\zeta-\gamma\xi,\quad \beta\xi-\alpha\eta.$$

11. To find the coordinates of the point O.

Consider O as the point of intersection of the tangents to the Hessian at the points E, F, then the coordinates of O are proportional to the terms of

$$\begin{vmatrix}3l^2X^2-\overline{1+2l^3}\,YZ, & 3l^2Y^2-\overline{1+2l^3}\,ZX, & 3l^2Z^2-\overline{1+2l^3}\,XY\\ 3l^2X'^2-\overline{1+2l^3}\,Y'Z', & 3l^2Y'^2-\overline{1+2l^3}\,Z'X', & 3l^2Z'^2-\overline{1+2l^3}\,X'Y'\end{vmatrix}.$$

Hence the x-coordinate is proportional to

$$(3l^2Y^2 - \overline{1+2l^3}\,ZX)(3l^2Z'^2 - \overline{1+2l^3}\,X'Y') - (3l^2Z^2 - \overline{1+2l^3}\,XY)(3l^2Y'^2 - \overline{1+2l^3}\,Z'X'),$$

which is equal to

$$9l^4(Y^2Z'^2-Y'^2Z^2)+3l^2(1+2l^3)\,YY'(XY'-X'Y)+3l^2(1+2l^3)\,ZZ'(ZX'-Z'X)-(1+2l^3)^2\,XX'(YZ'-Y'Z);$$

or introducing, as before, the quantities $\xi, \eta, \zeta, \alpha, \beta, \gamma$, to

$$\begin{aligned}&9l^4\alpha\xi + 3l^2(1+2l^3)(\beta\zeta+\gamma\eta) - (1+2l^3)^2\alpha\xi\\ &= (-1-13l^3+4l^6)\,\alpha\xi + 3l^2(1+2l^3)(\beta\zeta+\gamma\eta).\end{aligned}$$

But by the first of the equations (A) $\beta\zeta+\gamma\eta = 2l\alpha\xi$, and the preceding value thus becomes $(-1-7l^3+8l^6)\,\alpha\xi$. Hence throwing out the constant factor the coordinates of the point O are found to be

$$\alpha\xi, \quad \beta\eta, \quad \gamma\zeta.$$

12. The points G, O are conjugate poles of the cubic.

Take a, b, c for the coordinates of G, and a', b', c' for the coordinates of O; we have

$$\begin{aligned}a, b, c &= \gamma\eta - \beta\zeta,\ \alpha\zeta - \gamma\xi,\ \beta\xi - \alpha\eta,\\ a', b', c' &= \alpha\xi,\ \beta\eta,\ \gamma\zeta.\end{aligned}$$

These values give $aa' + l(bc' + b'c)$

$$\begin{aligned}&= \alpha\xi(\gamma\eta - \beta\zeta) + l\{(\alpha\zeta - \gamma\xi)\gamma\zeta + \beta\eta(\beta\xi - \alpha\eta)\}\\ &= \xi(\alpha\gamma\eta + l\beta^2\eta - l\alpha\eta^2) + \eta(\alpha\beta\gamma) + \zeta(-\alpha\beta\xi - l\gamma^2\xi)\end{aligned}$$

or substituting for $\xi\eta, \eta\zeta, \zeta\xi, \xi^2$ their values in terms of α, β, γ, this is

$$\begin{aligned}&\xi\{(-l\gamma^2 - \tfrac{1}{l}\gamma)(-4\gamma\alpha) + (-l\beta^2 - \tfrac{1}{l}\beta)(-l\alpha\beta)\}\\ &\quad + \eta\{(\alpha^2 l - \tfrac{4}{l}\alpha^2)\}\cdot(\alpha\beta\gamma)\\ &\quad + \zeta\{(-l\beta^2 - \tfrac{1}{l}\beta)(-\alpha\beta - l\gamma^2)\},\end{aligned}$$

which is identically equal to zero. Hence, completing the system, we find

$$\begin{aligned}aa' + l(bc' + b'c) &= 0,\\ bb' + l(ca' + c'a) &= 0,\\ cc' + l(ab' + a'b) &= 0,\end{aligned}$$

equations which show that O (as well as G) is a point of the Hessian, and that the points G, O are corresponding poles of the cubic.

13. The line EF joining a pair of conjugate poles of the cubic is a tangent of the Pippian[1].

In fact, the equations (A), by the elimination of α, β, γ, give

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1+4l^3)\,\xi\eta\zeta = 0,$$

which proves the theorem.

14. To find the equation of the pair of lines through F, and to show that these lines are tangents of the Pippian.

The equation of the pair of lines considered as the first or conic polar of the conjugate pole E, is

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) = 0.$$

[1]Steiner's curve R_3, in the particular case of a cubic basis-curve, is according to definition the envelope of the line EF; that is, the curve R_3 in the particular case in question is the Pippian.

Let one of the lines be $\lambda x + \mu y + \nu z = 0$, then the other is

$$\frac{X}{\lambda} + \frac{Y}{\mu} + \frac{Z}{\nu} = 0;$$

and we find

$$\begin{aligned} 2lX\mu\nu - Y\nu^2 - Z\mu^2 &= 0, \\ -X\nu^2 + 2lY\nu\lambda - Z\lambda^2 &= 0, \\ -X\mu^2 - Y\lambda^2 + 2lZ\lambda\mu &= 0, \end{aligned}$$

any two of which determine the ratios λ, μ, ν.

The elimination of X, Y, Z gives

$$\begin{vmatrix} 2l\mu\nu, & -\nu^2, & -\mu^2 \\ -\nu^2, & 2l\nu\lambda, & -\lambda^2 \\ -\mu^2, & -\lambda^2, & 2l\lambda\mu \end{vmatrix} = 0,$$

which is equivalent to

$$\lambda\mu\nu\{-l(\lambda^3 + \mu^3 + \nu^3) + (-1 + 4l^3)\,\lambda\mu\nu\} = 0;$$

or, omitting a factor, to

$$-l(\lambda^3 + \mu^3 + \nu^3) + (-1 + 4l^3)\,\lambda\mu\nu = 0,$$

which shows that the line in question is a tangent of the Pippian.

15. To find the equation of the pair of lines through O.

The equation of the pair of lines through E is in like manner

$$X'(x^2 + 2lyz) + Y'(y^2 + 2lzx) + Z'(z^2 + 2lxy) = 0;$$

and combining this with the foregoing equation,

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) = 0$$

of the pair of lines through F, viz. multiplying the two equations by

$$X'X' + Y'Y' + Z'Z', \quad -(XX' + YY' + ZZ'),$$

and adding, then if as before

$$a : b : c = \gamma\eta - \beta\zeta : \alpha\zeta - \gamma\xi : \beta\xi - \alpha\eta,$$

we find as the equation of a conic passing through the points A, B, C, D, the equation

$$a(x^2 + 2lyz) + b(y^2 + 2lzx) + c(z^2 + 2lxy) = 0.$$

But putting, as before,

$$a' : b' : c' = \alpha\xi : \beta\eta : \gamma\zeta,$$

then a', b', c' are the coordinates of the point O, and the equations

$$\begin{aligned} aa' + l(bc' + b'c) &= 0, \\ bb' + l(ca' + c'a) &= 0, \\ cc' + l(ab' + a'b) &= 0, \end{aligned}$$

show that the conic in question is in fact the pair of lines through the point O.

16. To find the coordinates of the point Γ, which is the harmonic of G with respect to the points E, F.

The coordinates of the point in question are

$$uX - vX',\ uY - vY',\ uZ - vZ',$$

where u, v have the values given in No. 10, viz.

$$u = -3l^2(XX'^2 + YY'^2 + ZZ'^2) + (1 + 2l^3)(Y'Z'X + Z'X'Y + X'Y'Z),$$
$$v = 3l^2(X^2X' + Y^2Y' + Z^2Z') - (1 + 2l^3)(YZX' + ZXY' + XYZ');$$

these values give

$$uX - vX' = -3l^2\{2X^2X'^2 + (XY' + X'Y)YY' + (XZ' + X'Z)ZZ'\} + (1 + 2l^3)\{(XY' + X'Y)(XZ' + X'Z) + XX'(YZ' + Y'Z)\};$$

and therefore

$$uX - vX' = -3l^2\{2\alpha^2 - \tfrac{2}{l}\beta\gamma\} + (1 + 2l^3)\big(\tfrac{1}{l^2}\beta\gamma - \tfrac{1}{l}\alpha^2\big) = \tfrac{1}{l}(1 + 8l^3)\big(-l\alpha^2 + \beta\gamma\big);$$

and consequently, omitting the constant factor, the coordinates of Γ may be taken to be

$$-l\alpha^2 + \beta\gamma, \quad -l\beta^2 + \gamma\alpha, \quad -l\gamma^2 + \alpha\beta.$$

17. The line through two consecutive positions of the point Γ is the line EF.

The coordinates of the point Γ are

$$-l\alpha^2 + \beta\gamma,\ -l\beta^2 + \gamma\alpha,\ -l\gamma^2 + \alpha\beta;$$

and it has been shown that the quantities α, β, γ satisfy the equation

$$-l(\alpha^3 + \beta^3 + \gamma^3) + (-1 + 4l^3)\,\alpha\beta\gamma = 0.$$

Hence, considering α, β, γ as variable parameters connected by this equation, the equation of the line through two consecutive positions of the point Γ is

$$\begin{vmatrix} -3l\alpha^2 + (-1 + 4l^3)\beta\gamma, & -3l\beta^2 + (-1 + 4l^3)\gamma\alpha, & -3l\gamma^2 + (-1 + 4l^3)\alpha\beta \\ x,\ -2l\alpha,\ \gamma,\ \beta \\ y,\ \gamma,\ -2l\beta,\ \alpha \\ z,\ \beta,\ \alpha,\ -2l\gamma \end{vmatrix} = 0,$$

and representing this equation by

$$Lx + My + Nz = 0,$$

we find

$$\begin{aligned} L = &-(4l^2\beta\gamma + \alpha^2)(-3l\alpha^2 + (-1 + 4l^3)\beta\gamma) \\ &+ (\alpha\beta + 2l\gamma^2)(-3l\beta^2 + (-1 + 4l^3)\gamma\alpha) \\ &+ (\alpha\gamma + 2l\beta^2)(-3l\gamma^2 + (-1 + 4l^3)\alpha\beta); \end{aligned}$$

or, multiplying out and collecting,

$$L = 3l\alpha^4 + (-1 - 8l^3)\alpha^2\beta\gamma + (-5l + 14l^4)(\alpha\beta^3 + \alpha\gamma^3) + (-16l^2 + 16l^5)\beta^2\gamma^2;$$

but the equation

$$-l(\alpha^3 + \beta^3 + \gamma^3) + (-1 + 4l^3)\,\alpha\beta\gamma = 0$$

gives

$$3l\alpha^4 = -3l(\alpha\beta^3 + \alpha\gamma^3) + (-3 + 12l^3)\alpha^2\beta\gamma,$$

and we have

$$\begin{aligned} L &= (-4 + 4l^3)\,\alpha^2\beta\gamma + (-8l + 8l^4)(\alpha\beta^3 + \alpha\gamma^3) + (-16l^2 + 16l^5)\beta^2\gamma^2 \\ &= (-4 + 4l^3)(\alpha^2\beta\gamma + 2l(\alpha\beta^3 + \alpha\gamma^3) + 4l^2\beta^2\gamma^2) \\ &= (-4 + 4l^3)(\alpha\gamma + 2l\beta^2)(\alpha\beta + 2l\gamma^2); \end{aligned}$$

or, in virtue of the equations (D),

$$L = (-4 + 4l^3) \cdot l^2\eta \cdot l^2\zeta = (-4 + 4l^3)\, l^4\eta\zeta.$$

Hence, omitting the common factor, we find $L : M : N = \xi : \eta : \zeta$, and the equation $Lx + My + Nz = 0$ becomes

$$\xi x + \eta y + \zeta z = 0,$$

which is the equation of the line EF, that is, the line through two consecutive positions of Γ is the line EF; or what is the same thing, the line EF touches the Pippian in the point Γ which is the harmonic of G with respect to the points E, F.

18. The lineo-polar envelope of the line EF, with respect to the cubic, is the pair of lines OE, OF.

The equation of the pair of lines OE, OF, considered as the tangents to the Hessian at the points E, F, is

$$\{(3l^2X^2 - \overline{1+2l^3}\,YZ)x + (3l^2Y^2 - \overline{1+2l^3}\,ZX)y + (3l^2Z^2 - \overline{1+2l^3}\,XY)z\} \times \{(3l^2X'^2 - \overline{1+2l^3}\,Y'Z')x + (3l^2Y'^2 - \overline{1+2l^3}\,Z'X')y + (3l^2Z'^2 - \overline{1+2l^3}\,X'Y')z\} = 0.$$

Here on the left-hand side the coefficient of x^2 is

$$9l^4X^2X'^2 - 3l^2(1+2l^3)(X^2Y'Z' + X'^2YZ) + (1+2l^3)^2YY'ZZ',$$

which is equal to

$$9l^4\alpha^2 - 3l^2(1+2l^3)\big(l^2\beta\gamma + \tfrac{1}{l}\alpha^2\big) + (1+2l^3)^2\beta\gamma,$$

that is

$$\tfrac{1}{l}(-l+l^4)\big\{3l\alpha^2 + 2(1+2l^3)\beta\gamma\big\};$$

and the coefficient of yz is

$$9l^4(Y^2Z'^2 + Y'^2Z^2) - 3l^2(1+2l^3)\big(YY'(XY' + X'Y) + ZZ'(XZ' + X'Z)\big) + (1+2l^3)^2XX'(YZ' + Y'Z),$$

which is equal to

$$9l^2\big(\tfrac{1}{l^4}\eta^2 - 2l\beta\gamma\big) - 3l^2(1+2l^3)\big(-\tfrac{2}{l}\beta\gamma\big) + (1+2l^3)^2\alpha\big(-\tfrac{1}{l}\alpha\big),$$

that is

$$\tfrac{1}{l}(-l+l^4)\big\{(1-4l^3)\alpha^2 - 6l^2\beta\gamma\big\}.$$

Hence completing the system and throwing out the constant factor, the equation of the pair of lines is

$$\big(3l\alpha^2 + 2(1+2l^3)\beta\gamma,\ 3l\beta^2 + 2(1+2l^3)\gamma\alpha,\ 3l\gamma^2 + 2(1+2l^3)\alpha\beta,\ (1-4l^3)\alpha^2 - 6l^2\beta\gamma,\ (1-4l^3)\beta^2 - 6l^2\gamma\alpha,\ (1-4l^3)\gamma^2 - 6l^2\alpha\beta \mid x, y, z\big)^2 = 0.$$

But the equation of the line EF is $\xi x + \eta y + \zeta z = 0$, and the equation of its lineo-polar envelope is

$$\begin{vmatrix} \xi, & x, & lz, & ly \\ \eta, & lz, & y, & lx \\ \zeta, & ly, & lx, & z \end{vmatrix} = 0;$$

or expanding,

$$\big(\xi z - l^2x^2,\ zx - l^2y^2,\ xy - l^2z^2,\ l^2yz - lx^2,\ l^2zx - ly^2,\ l^2xy - lz^2 \mid \xi, \eta, \zeta\big) = 0;$$

or arranging in powers of x, y, z,

$$\big(-l\xi^2,\ -l\eta^2 - 2l\zeta\xi,\ -l\zeta^2 - 2l\xi\eta,\ -l\xi^2 - 2l\eta\zeta,\ l^2\eta^2 + l^2\zeta\xi,\ l^2\zeta^2 + l^2\xi\eta \mid x, y, z\big)^2 = 0;$$

and if in this equation we replace ξ^2 &c. by their values in terms of α, β, γ, as given by the equations (D), we obtain the equation given as that of the pair of lines OE, OF.

19. It remains to prove the theorem with respect to the connexion of the lines EF, IJ.

The equations (A) show that the two lines

$$\xi x + \eta y + \zeta z = 0,$$
$$\alpha x + \beta y + \gamma z = 0,$$

(where ξ, η, ζ and α, β, γ have the values before attributed to them) are conjugate polars with respect to the curve of the third class,

$$(\,2l\xi,\ \zeta,\ \eta,\ -l\xi^2,\ \zeta x,\ \eta y\,)\ldots\ldots = 0,$$

in which equation ξ, η, ζ denote current line coordinates. The curve in question is of the form

$$A \cdot PU + B \cdot QU = 0.$$

We have, in fact, identically,

$$3T \cdot PU - 4S \cdot QU = (1 + 8l^3)^2\{l(\xi^3 + \eta^3 + \zeta^3) - 3\xi\eta\zeta\}.$$

It is clear that the curve in question must have the curve $PU = 0$ for its Hessian; and in fact, in the formula of my Third Memoir, [144]

$$H(6aPU + 6\beta QU) = (-2T,\ 4S^2,\ 18TS,\ T^2 + 16S^3 \mid \alpha, \beta)^3 PU + (8S,\ T,\ -8S^2,\ -TS \mid \alpha, \beta)^3 QU.$$

The coefficient of QU is

$$(8S,\ T,\ -8S^2,\ -TS \mid \alpha, \beta)^3,$$

and therefore, putting $\alpha = \frac{1}{3}T$, $\beta = -\frac{4}{3}S$, we find

$$H(3T \cdot PU - 4S \cdot QU) = -l(T^2 - 64S^3)^2\, PU.$$

Article No. 20. Theorem relating to the curve of the third class, mentioned in the preceding Article.

20. The consideration of the curve $3T \cdot PU - 4S \cdot QU = 0$, gives rise to another geometrical theorem. Suppose that the line (ξ, η, ζ), that is, the line whose equation is $\xi x + \eta y + \zeta z = 0$, is with respect to this curve of the third class one of the four polars of a point (X, Y, Z) of the Hessian, and that it is required to find the envelope of the line $\xi x + \eta y + \zeta z = 0$.

We have

$$X : Y : Z = l\xi^2 - \eta\zeta\ :\ l\eta^2 - \zeta\xi\ :\ l\zeta^2 - \xi\eta,$$

and X, Y, Z are to be eliminated from these equations, and the equation

$$l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ = 0$$

of the Hessian. We have

$$X^3 + Y^3 + Z^3 = l^3(\xi^3 + \eta^3 + \zeta^3)^2,$$

$$XYZ = -l^2(\xi^3 + \eta^3 + \zeta^3)\,\xi\eta\zeta,$$

and thence

$$HU = l^4(\xi^3 + \eta^3 + \zeta^3)^2 \cdot \{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\};$$

and equating the right-hand side to zero, we have the equation in line coordinates of the curve in question, which is therefore a curve of the sixth class in quadratic syzygy with the Pippian and Quippian.

Article No. 21. Geometrical definition of the Quippian.

21. I have not succeeded in obtaining any good geometrical definition of the Quippian, and the following is only given for want of something better.

The curve

$$T \cdot PU \cdot \{P(6H(aU + 6\beta HU))\} - P(6HU)\{T(aU + 6\beta HU) \cdot P(aU + 6\beta HU)\} = 0,$$

which is derived in what may be taken to be a known manner from the cubic, is in general a curve of the sixth class. But if the syzygetic cubic $aU + 6\beta HU = 0$ be properly selected, viz. if this curve be such that its Hessian breaks up into three lines, then both the Pippian of the cubic $aU + 6\beta HU = 0$, and the Pippian of its Hessian will break up into the same three points, which will be a portion of the curve of the sixth class, and discarding these three points the curve will sink down to one of the third class, and will in fact be the Quippian of the cubic.

To show this we may take

$$aU + 6\beta HU = x^3 + y^3 + z^3$$

as the equation of the syzygetic cubic satisfying the prescribed condition, for this value in fact gives

$$H(aU + 6\beta HU) = -xyz,$$

a system of three lines. We find, moreover,

$$P(aU + 6\beta HU) = -\xi\eta\zeta,$$

and

$$P\{6H(aU + 6\beta HU)\} = P(-6xyz) = -4\xi\eta\zeta,$$

the latter equation being obtained by first neglecting all but the highest power of l in the expression of PU, and then writing $l = -1$; we have also $T(aU + 6\beta HU) = 1$. Substituting the above values, the curve of the sixth class is

$$\xi^2\eta^2\zeta^2\{-4T \cdot PU + P(6HU)\} = 0;$$

or throwing out the factor $\xi^2\eta^2\zeta^2$, we have the curve of the third class,

$$-4T \cdot PU + P(6HU) = 0.$$

Now the general expression in my Third Memoir, viz.

$$P(aU + 6\beta HU) = \{(\alpha^3 + l^2)\alpha\beta^2 + 4T\beta^3\}PU + (\alpha^4 - 4S\beta^4)QU,$$

putting $\alpha = 0$, $\beta = 1$, gives

$$P(6HU) = 4T \cdot PU - 4S \cdot QU,$$

or what is the same thing,

$$-4T \cdot PU + P(6HU) = -4S \cdot QU;$$

and the curve of the third class is therefore the Quippian $QU = 0$. It may be remarked, that for a cubic $U = 0$ the Hessian of which breaks up into three lines, the above investigation shows that we have $PU = -\xi\eta\zeta$, $P(6HU) = -4\xi\eta\zeta$, and $T = 1$, and consequently that $-4T \cdot PU + P(6HU)$ ought to vanish identically; this in fact happens in virtue of the factor S on the right-hand side, the invariant S of a cubic of the form in question being equal to zero; the appearance of the factor S on the right-hand side is thus accounted for *à priori*.

Article No. 22. Theorem relating to a line which meets three given conics in six points in involution.

22. The envelope of a line which meets three given conics, the first or conic polars of any three points with respect to the cubic, in six points in involution, is the Pippian.

It is readily seen that if the theorem is true with respect to the three conics,

$$\frac{dU}{dx} = 0, \quad \frac{dU}{dy} = 0, \quad \frac{dU}{dz} = 0,$$

it is true with respect to any three conics whatever of the form

$$\lambda\frac{dU}{dx} + \mu\frac{dU}{dy} + \nu\frac{dU}{dz} = 0,$$

that is, with respect to any three conics, each of them the first or conic polar of some point (λ, μ, ν) with respect to the cubic. Considering then these three conics, take $\xi x + \eta y + \zeta z = 0$ as the equation of the line, and let (X, Y, Z) be the coordinates of a point of intersection with the first conic, we have

$$\xi X + \eta Y + \zeta Z = 0,$$
$$X^2 + 2lYZ = 0;$$

and combining with these a linear equation

$$\alpha X + \beta Y + \gamma Z = 0,$$

in which (α, β, γ) are arbitrary quantities, we have

$$X : Y : Z = \beta\zeta - \gamma\eta \, : \, \gamma\xi - \alpha\zeta \, : \, \alpha\eta - \beta\xi;$$

and hence

$$(\beta\zeta - \gamma\eta)^2 + 2l(\gamma\xi - \alpha\zeta)(\alpha\eta - \beta\xi) = 0,$$

an equation in (α, β, γ) which is in fact the equation in line coordinates of the two points of intersection with the first conic. Developing and forming the analogous equations, we find

$$\begin{aligned}
&\big(-2l\eta\zeta,\ \zeta^2,\ \eta^2,\ -\eta\zeta - l\xi^2,\ l\eta\xi,\ l\xi\zeta \mid \alpha, \beta, \gamma\big)^2 = 0,\\
&\big(\zeta^2,\ -2l\zeta\xi,\ \xi^2,\ l\xi\eta,\ -l\eta^2 - l\xi^2,\ l\eta\zeta \mid \alpha, \beta, \gamma\big)^2 = 0,\\
&\big(\eta^2,\ \xi^2,\ -2l\xi\eta,\ l\eta\zeta,\ l\xi\zeta,\ -\xi\eta - l\zeta^2 \mid \alpha, \beta, \gamma\big)^2 = 0,
\end{aligned}$$

which are respectively the equations in line coordinates of the three pairs of intersections.

Now combining these equations with the equation $\gamma = 0$, we have the equations of the pairs of lines joining the points of intersection with the point $(x = 0, y = 0)$, and if the six points are in involution, the six lines must also be in involution, or the condition for the involution of the six points is

$$\begin{vmatrix} -2l\eta\zeta, & \zeta^2, & l\xi\zeta \\ \zeta^2, & -2l\zeta\xi, & l\xi\eta \\ \eta^2, & \xi^2, & l\eta\zeta \end{vmatrix} = 0,$$

that is,

$$4l^2\xi^2\xi\eta(-\xi\eta - l\zeta^2) + l\xi^2\eta\zeta(-\zeta^2 - l\xi^2) + l^2\xi\eta\zeta^4 + 2l^2\xi^2\eta\zeta^3 + 2l^2\xi^2\eta\zeta^3 + \zeta^4(-l^3 - l^4) = 0;$$

or, reducing and throwing out the factor ζ, we find

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\,\xi\eta\zeta = 0,$$

which shows that the line in question is a tangent of the Pippian.

It is to be remarked that any three conics whatever may be considered as the first or conic polars of three properly selected points with respect to a properly selected cubic curve. The theorem applies therefore to any three conics whatever, but in this case the cubic curve is not given, and the Pippian therefore stands merely for a curve of the third class, and the theorem is as follows, viz. the envelope of a line which meets any three conics in six points in involution, is a curve of the third class.

Article No. 23. Completion of the theory in Liouville, and comparison with analogous theorems of Hesse.

In order to convert the foregoing theorem into its reciprocal, we must replace the cubic $U = 0$ by a curve of the third class, that is we must consider the coordinates which enter into the equation as line coordinates; and it of course follows that the coordinates which enter into the equation $PU = 0$ must be considered as point coordinates, that is we must consider the Pippian as a curve of the third order: we have thus the theorem; The locus of a point such that the tangents drawn from it to three given conics (the first or conic poles of any three lines with respect to a curve of the third class) form a pencil in involution, is the Pippian considered as a curve of the third order. This in fact completes the fundamental theorem in my memoirs in Liouville above referred to, and establishes the analogy with Hesse's results in relation to the Hessian; to show this I set out the two series of theorems as follows.

Hesse, in his memoirs On Curves of the Third Order and Curves of the Third Class, Crelle, tt. XXVIII., XXXVI., and XXXVIII. [1844, 1848, 1849], has shown as follows:

(α) The locus of a point such that its polars with respect to the three conics $X = 0$, $Y = 0$, $Z = 0$ (or more generally its polars with respect to all the conics of the series $\lambda X + \mu Y + \nu Z = 0$) meet in a point, is a curve of the third order $V = 0$.

(β) Conversely, given a curve of the third order $V = 0$, there exists a series of conics such that the polars with respect to all the conics of any point whatever of the curve $V = 0$, meet in a point.

(γ) The equation of any one of the conics in question is

$$\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz} = 0,$$

that is, the conic is the first or conic polar of a point (λ, μ, ν) with respect to a certain curve of the third order $U = 0$; and this curve is determined by the condition that its Hessian is the given curve $V = 0$, that is, we have $V = HU$.

(δ) The equation $V = HU$ is solved by assuming $U = aV + bHV$, for we have then $H(aV + bHV) = AV + BHV$, where A, B are given cubic functions of a, b, and thence $V = HU = AV + BHV$, or $A = 1$, $B = 0$; the latter equation gives what is alone important, the ratio $a : b$; and it thus appears that there are three distinct series of conics, each of them having the above-mentioned relation to the given curve of the third order $V = 0$.

In the memoirs in Liouville above referred to, I have in effect shown that

(α') The locus of a point such that the tangents from it to three conics, represented in line coordinates by the equations $X = 0$, $Y = 0$, $Z = 0$ (or more generally with respect to any three conics of the series $\lambda X + \mu Y + \nu Z = 0$) form a pencil in involution, is a curve of the third order $V = 0$.

(β') Conversely, given a curve of the third order $V = 0$, there exists a series of conics such that the tangents from any point whatever of the curve to any three of the conics, form a pencil in involution.

Now, considering the coordinates which enter into the equation of the Pippian as point coordinates, and consequently the Pippian as a curve of the third order, I am able to add as follows:

(γ') The equation in line coordinates of any one of the conics in question is

$$\lambda \frac{dU}{d\xi} + \mu \frac{dU}{d\eta} + \nu \frac{dU}{d\zeta} = 0,$$

that is, the conic is the first or conic polar of a line (λ, μ, ν) with respect to a certain curve of the third class $U = 0$; and this curve is determined by the condition that its Pippian is the given curve of the third order $V = 0$, that is, we have $V = PU$.

(δ') The equation $V = PU$ is solved by assuming $U = aPV + bQV$, for we have then $P(aPV + bQV) = AV + BHV$, where A and B are given cubic functions of a, b; and thence $V = PU = AV + BHV$, or $A = 1$, $B = 0$; the latter equation gives what is alone important, the ratio $a : b$; and it thus appears that there are three distinct curves of the third class $U = 0$, and therefore (what indeed is shown in the Memoirs in Liouville) three distinct series of conics having the above-mentioned relation to the given curve of the third order $V = 0$.

It is hardly necessary to remark that the preceding theorems, although precisely analogous to those of Hesse, are entirely distinct theorems, that is the two series are not connected together by any relation of reciprocity.

Article Nos. 24 to 28. Various investigations and theorems.

24. Reverting to the theorem (No. 18), that the lineo-polar envelope of the line EF is the pair of lines OE, OF; the line EF is any tangent of the Pippian, hence the theorem includes the following one:

The lineo-polar envelope with respect to the cubic, of any tangent of the Pippian, is a pair of lines.

And conversely,

The Pippian is the envelope of a line such that the lineo-polar envelope of the line with respect to the cubic is a pair of lines.

It is I think worth while to give an independent proof. It has been shown that the equation of the lineo-polar envelope with respect to the cubic, of the line $\xi x + \eta y + \zeta z = 0$ (where ξ, η, ζ are arbitrary quantities), is

$$\left(-l\xi^2,\ -l\eta^2 - 2l\zeta\xi,\ -l\zeta^2 - 2l\xi\eta,\ -l\xi^2 - 2l\eta\zeta,\ l^2\eta^2 + l^2\zeta\xi,\ l^2\zeta^2 + l^2\xi\eta \mid x, y, z\right)^2 = 0;$$

and representing this equation by

$$\tfrac{1}{l}\left(a, b, c, f, g, h \mid x, y, z\right)^2 = 0,$$

we find

$$\begin{aligned}
bc - f^2 &= l^2(8l^3\xi^4 - \eta^4 + 4l\eta\zeta^3 + 4l^2\zeta^2\eta^2\xi),\\
ca - g^2 &= l^2(8l^3\eta^4 + 8l\zeta^3\xi - \zeta^4 + 12l^2\zeta^2\xi^2\eta),\\
ab - h^2 &= l^2(8l^3\zeta^4 + 12l^2\xi^2\eta^2 + \ldots),\\
gh - af &= \xi\{2l^3(\xi^4+\eta^4+\zeta^4) + 4l(1+2l^3)\xi\eta\zeta\} + (1+8l^3)\eta^2\zeta^2,\\
hf - bg &= \eta\{2l^3(\xi^4+\eta^4+\zeta^4) + 4l(1+2l^3)\xi\eta\zeta\} + (1+8l^3)\zeta^2\xi^2,\\
fg - ch &= \zeta\{2l^3(\xi^4+\eta^4+\zeta^4) + 4l(1+2l^3)\xi\eta\zeta\} + (1+8l^3)\xi^2\eta^2;
\end{aligned}$$

and after all reductions,

$$abc - af^2 - bg^2 - ch^2 + 2fgh = -l(\xi^3+\eta^3+\zeta^3)\{-l(\xi^3+\eta^3+\zeta^3) + (-1+4l^3)\xi\eta\zeta\}^2 = -l\,(PU)^2,$$

or the condition in order that the conic may break up into a pair of lines is $PU = 0$.

25. The following formulae are given in connexion with the foregoing investigation, but I have not particularly considered their geometrical signification. The lineo-polar envelope of an arbitrary line $\xi x+\eta y+\zeta z = 0$, with respect to the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

has been represented by

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0;$$

and if in like manner we represent the lineo-polar envelope of the same line, with respect to a syzygetic cubic

$$x^3 + y^3 + z^3 + 6l'xyz = 0,$$

by

$$(a', b', c', f', g', h' \mid x, y, z)^2 = 0,$$

then we have

$$\begin{aligned}
&a'(bc - f^2) + b'(ca - g^2) + c'(ab - h^2) + 2f'(gh - af) + 2g'(hf - bg) + 2h'(fg - ch)\\
&\quad= \big(l^4(\xi^6+\eta^6+\zeta^6) + (2l^2 + 4l - 32l^3l'^2 + 8l^6)(\xi^3+\eta^3+\zeta^3)\xi\eta\zeta\\
&\qquad+ (24l^2l'^2 + 48l^4l'^2 - 11ll' + 24l^7 + a)\,\xi^2\eta^2\zeta^2\big),
\end{aligned}$$

which may be verified by writing $l' = l$, in which case the right-hand side becomes as it should do, $S(PU)^2$. If $l' = \dfrac{1+2l^3}{-6l^2}$, that is, if the syzygetic cubic be the Hessian, then the formula becomes

$$a'(bc-f^2)+\&c. = 3l^4(\xi^6+\eta^6+\zeta^6)+12l^2(-1+26l^3+56l^6)(\xi^3+\eta^3+\zeta^3)\xi\eta\zeta+12l(2+57l^3+168l^6+16l^9)\,\xi^2\eta^2\zeta^2,$$

which is equal to

$$\frac{T^2\cdot 25S - 24S\cdot PU}{36l^3}.$$

26. The equation

$$(bc' + b'c - 2ff',\ \ldots,\ gh' + g'h - af - a'f',\ \ldots \mid \xi, \eta, \zeta)^2 = 0$$

is the equation in line coordinates of a conic, the envelope of the line which cuts harmonically the conics

$$\begin{aligned}
(a, b, c, f, g, h \mid x, y, z)^2 &= 0,\\
(a', b', c', f', g', h' \mid x, y, z)^2 &= 0;
\end{aligned}$$

and if a, b, &c., a', &c. have the values before given to them, then the coefficients of the equation are

$$\begin{aligned}
bc' + b'c - 2ff' &= \xi\{-\xi^3 + 4ll'(l+l')(\eta^3+\zeta^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\,\xi\eta\zeta\},\\
ca' + c'a - 2gg' &= \eta\{-\eta^3 + 4ll'(l+l')(\zeta^3+\xi^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\,\xi\eta\zeta\},\\
ab' + a'b - 2hh' &= \zeta\{-\zeta^3 + 4ll'(l+l')(\xi^3+\eta^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\,\xi\eta\zeta\},\\
gh' + g'h - af - a'f' &= \xi\{(l^3+l'^3)(\xi^3+\eta^3+\zeta^3) + (2l+2l'+8l^2l'^2)\xi\eta\zeta\} + (1+4ll'(l+l'))\,\eta^2\zeta^2,\\
hf' + h'f - bg - b'g' &= \eta\{(l^3+l'^3)(\xi^3+\eta^3+\zeta^3) + (2l+2l'+8l^2l'^2)\xi\eta\zeta\} + (1+4ll'(l+l'))\,\zeta^2\xi^2,\\
fg' + f'g - ch - c'h' &= \zeta\{(l^3+l'^3)(\xi^3+\eta^3+\zeta^3) + (2l+2l'+8l^2l'^2)\xi\eta\zeta\} + (1+4ll'(l+l'))\,\xi^2\eta^2;
\end{aligned}$$

and we thence obtain

$$\tfrac{1}{ll'}(bc'+b'c-2ff',\ \ldots,\ gh'+g'h-af-a'f',\ \ldots\,|\,\xi,\eta,\zeta)^2 = -(\xi^4+\eta^4+\zeta^4)+(l^2+l'^2+1ll')(\xi^3+\eta^3+\zeta^3)\xi\eta\zeta+(6l+6l'+24l^2l'^2)\,\xi^2\eta^2\zeta^2+(4+16(l+l')(l^2+l'^2))(y^2\zeta^2+\zeta^2\xi^2+\xi^2\eta^2)=0$$

as the condition which expresses that a line $\xi x+\eta y+\zeta z=0$ cuts harmonically its lineo-polar envelopes with respect to the cubic and with respect to a syzygetic cubic.

27. To find the locus of a point such that its second or line polar with respect to the cubic may be a tangent of the Pippian. Let the coordinates of the point be (x,y,z); then if $\xi x+\eta y+\zeta z=0$ be the equation of the polar, we have

$$\xi:\eta:\zeta = x^2+2lyz\ :\ y^2+2lzx\ :\ z^2+2lxy,$$

and the line in question being a tangent to the Pippian,

$$-l(\xi^3+\eta^3+\zeta^3)+(-1+4l^3)\,\xi\eta\zeta=0.$$

But the preceding values give

$$\begin{aligned}\xi^3+\eta^3+\zeta^3 &= x^6+y^6+z^6+(2+8l^3)(y^3z^3+z^3x^3+x^3y^3)+6l\cdot 6(x^3+y^3+z^3)xyz,\\ \xi\eta\zeta &= x^2y^2z^2+2l^3(x^3+y^3+z^3)xyz+(1+8l^3)x^2y^2z^2+2l(y^3z^3+z^3x^3+x^3y^3);\end{aligned}$$

and we have therefore

$$-l(x^6+y^6+z^6)-(10l^3+16l^4)(x^3+y^3+z^3)xyz+(1+40l^3+32l^6)x^2y^2z^2-\ldots=0;$$

or introducing U, HU in place of $x^3+y^3+z^3$, xyz, the equation becomes

$$-S\cdot U^2+(HU)^2=0,$$

which is the equation of the locus in question.

28. The locus of a point such that its second or line polar with respect to the cubic is a tangent of the Quippian, is found in like manner by substituting the last-mentioned values of ξ,η,ζ in the equation

$$QU=(1-10l^3)(\xi^3+\eta^3+\zeta^3)-6l^2(5+4l^3)\,\xi\eta\zeta.$$

We find as the equation of the locus,

$$(1-10l^3)(x^6+y^6+z^6)-4l(1-30l^3-16l^6)(x^3+y^3+z^3)xyz+6l^2(1-104l^3-32l^6)x^2y^2z^2-2(1+8l^3)^2(y^3z^3+z^3x^3+x^3y^3)=0,$$

where the function on the left-hand side is the octicovariant $\Theta_{14}U$ of my Third Memoir, the covariant having been in fact defined so as to satisfy the condition in question. And I have given in the memoir the following expression for $\Theta_{14}U$, viz.

$$\begin{aligned}\Theta_{14}U = {} & (1-16l^3-6l^6)U^3\\ & +(6l^2\cdot\ldots)U\cdot HU\\ & +(6l^2\cdot\ldots)(HU)^2\\ & -2(1+8l^3)^2(y^3z^3+z^3x^3+x^3y^3)^?.\end{aligned}$$

Article Nos. 29 to 31. Formulae for the intersection of a cubic curve and a line.

29. If the line $\xi x+\eta y+\zeta z=0$ meet the cubic

$$x^3+y^3+z^3+6lxyz=0$$

in the points

$$(x_1,y_1,z_1),\ (x_2,y_2,z_2),\ (x_3,y_3,z_3),$$

then we have

$$x_1x_2x_3\ :\ y_1y_2y_3\ :\ z_1z_2z_3=\eta^3-\zeta^3\ :\ \zeta^3-\xi^3\ :\ \xi^3-\eta^3.$$

It will be convenient to represent the equation of the cubic by the abbreviated notation $(1,1,1,l \mid x,y,z)^3 = 0$; we have the two equations

$$\begin{aligned}(1,1,1,l \mid x,y,z)^3 &= 0,\\ \xi x + \eta y + \zeta z &= 0;\end{aligned}$$

and if to these we join a linear equation with arbitrary coefficients,

$$\alpha x + \beta y + \gamma z = 0,$$

then the second and third equations give

$$x : y : z = \beta\zeta - \gamma\eta \,:\, \gamma\xi - \alpha\zeta \,:\, \alpha\eta - \beta\xi,$$

and substituting these values in the first equation, we obtain the resultant of the system. But this resultant will also be obtained by substituting, in the third equation, a system of simultaneous roots of the first and second equations, and equating to zero the product of the functions so obtained[2]. We must have therefore

$$(1,1,1,l \mid \beta\zeta - \gamma\eta,\ \gamma\xi - \alpha\zeta,\ \alpha\eta - \beta\xi)^3 = (\alpha x_1 + \beta y_1 + \gamma z_1)(\alpha x_2 + \beta y_2 + \gamma z_2)(\alpha x_3 + \beta y_3 + \gamma z_3),$$

and equating the coefficients of $\alpha^3, \beta^3, \gamma^3$, we obtain the above-mentioned relations.

30. If a tangent to the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0$$

at a point (x_1, y_1, z_1) of the cubic meet the cubic in the point (x_2, y_2, z_2), then

$$x_2 : y_2 : z_2 = x_1(y_1^3 - z_1^3) \,:\, y_1(z_1^3 - x_1^3) \,:\, z_1(x_1^3 - y_1^3).$$

For if the equation of the tangent is $\xi x + \eta y + \zeta z = 0$, then

$$\xi^3 - \eta^3 = x_1^3 x_2 \,:\, y_1^3 y_2 \,:\, z_1^3 z_2 \,:\, \xi^2,\ \eta^2,\ \zeta^2,$$

and

$$\xi : \eta : \zeta = x_1^2 + 2ly_1z_1 \,:\, y_1^2 + 2lz_1x_1 \,:\, z_1^2 + 2lx_1y_1.$$

These values give

$$\xi^3 - \eta^3 = \dots$$

since (x_1, y_1, z_1) is a point of the cubic; and forming in like manner the values of $\xi^3 - \zeta^3$ and $\zeta^3 - \eta^3$, we obtain the theorem.

31. The preceding values of (x_2, y_2, z_2) ought to satisfy

$$\begin{aligned}&(x_1^2 + 2ly_1z_1)x_2 + (y_1^2 + 2lz_1x_1)y_2 + (z_1^2 + 2lx_1y_1)z_2 = 0,\\ &x_2^3 + y_2^3 + z_2^3 + 6lx_2y_2z_2 = 0;\end{aligned}$$

in fact the first equation is satisfied identically, and for the second equation we obtain

$$\begin{aligned}x_2^3 + y_2^3 + z_2^3 &= x_1^3(y_1^3 - z_1^3)^3 + y_1^3(z_1^3 - x_1^3)^3 + z_1^3(x_1^3 - y_1^3)^3\\ &= -x_1^3(y_1^3 - z_1^3)\left[y_1^3(z_1^3 - x_1^3) \cdot z_1^3(x_1^3 - y_1^3)\right] \cdot \dots\\ &= -x_1y_1z_1 \cdot x_1y_1z_1(y_1^3 - z_1^3)(z_1^3 - x_1^3)(x_1^3 - y_1^3),\end{aligned}$$

and consequently

$$x_2^3 + y_2^3 + z_2^3 + 6lx_2y_2z_2 = (x_1^3 + y_1^3 + z_1^3 + 6lx_1y_1z_1)(y_1^3 - z_1^3)(z_1^3 - x_1^3)(x_1^3 - y_1^3) = 0,$$

which verifies the theorem. It is proper to add (the remark was made to me by Professor Sylvester) that the foregoing values

$$x_2 : y_2 : z_2 = x_1(y_1^3 - z_1^3) \,:\, y_1(z_1^3 - x_1^3) \,:\, z_1(x_1^3 - y_1^3)$$

[2] This is in fact the general process of elimination given in Schläfli's Memoir, "Ueber die Resultante eines Systemes mehrerer algebraischer Gleichungen," Vienna Trans. 1852. [But the process was employed much earlier, by Poisson.]

satisfy identically the relation
$$\frac{x_2^3+y_2^3+z_2^3}{x_2y_2z_2}=\frac{x_1^3+y_1^3+z_1^3}{x_1y_1z_1}.$$

Article Nos. 32 to 34. Formulae for the Satellite line and point.

32. The line $\xi x+\eta y+\zeta z=0$ meets the cubic
$$x^3+y^3+z^3+6lxyz=0$$
in three points, and the tangents to the cubic at these points meet the cubic in three points lying in a line, which has been called the Satellite line of the given line.

To find the equation of the satellite line; suppose that (x_1,y_1,z_1), (x_2,y_2,z_2), (x_3,y_3,z_3) are the coordinates of the points in which the given line meets the cubic; then we have, as before,
$$\big(1,1,1,l\mid\beta\zeta-\gamma\eta,\ \gamma\xi-\alpha\zeta,\ \alpha\eta-\beta\xi\big)^3=(\alpha x_1+\beta y_1+\gamma z_1)(\alpha x_2+\beta y_2+\gamma z_2)(\alpha x_3+\beta y_3+\gamma z_3).$$

The equation of the three tangents is
$$\begin{aligned}\Pi=&\big[(x_1^2+2ly_1z_1)x+(y_1^2+2lz_1x_1)y+(z_1^2+2lx_1y_1)z\big]\\ &\times\big[(x_2^2+2ly_2z_2)x+(y_2^2+2lz_2x_2)y+(z_2^2+2lx_2y_2)z\big]\\ &\times\big[(x_3^2+2ly_3z_3)x+(y_3^2+2lz_3x_3)y+(z_3^2+2lx_3y_3)z\big]=0,\end{aligned}$$
and if we put
$$F=(\xi^3+\eta^3+\zeta^3)^2-2l^3(\xi^3+\eta^3+\zeta^3)\,\xi\eta\zeta+(-2l-l^4)\,\xi^3\eta^3\zeta^3+(-4+32l^3)(\eta^3\zeta^3+\zeta^3\xi^3+\xi^3\eta^3),$$
(F is the reciprocant FU of my Third Memoir), then we have identically
$$F\cdot U-\Pi=(\xi x+\eta y+\zeta z)^2(\xi' x+\eta' y+\zeta' z),$$
and the equation of the satellite line is $\xi' x+\eta' y+\zeta' z=0$. In fact the geometrical theory shows that we must have
$$F\cdot U-N\,\Pi=(\xi x+\eta y+\zeta z)^2(\xi' x+\eta' y+\zeta' z),$$
and it is then clear that N is a mere number. To determine its value in the most simple manner, write $x=1,\ y=0,\ z=-\dfrac{\xi}{\zeta}$, we have then $F\cdot U-N\,\Pi=0$, where
$$F=\ldots.$$
The value of Π is $\Pi=F\cdot U$, and we thus obtain $N=1$. For, substituting the above values,
$$\begin{aligned}\Pi=&(x_1^2\zeta-z_1^2\xi)(x_2^2\zeta-z_2^2\xi)(x_3^2\zeta-z_3^2\xi)\\ &-\zeta^3x_1x_2x_3z_1z_2z_3\cdot\ldots\\ &-\zeta^2\xi(x_1^2z_2^2z_3^2+\&\text{c.})\\ &+\zeta\xi^2(x_1^2z_2^2+\&\text{c.})\\ &-\xi^3z_1z_2z_3,\end{aligned}$$
and we have
$$\begin{aligned}x_1x_2x_3&=\eta^3-\zeta^3,\\ x_1x_2z_3+\&\text{c.}&=3\xi^2\zeta,\\ x_1z_2z_3+\&\text{c.}&=-3\xi\zeta^2,\\ z_1z_2z_3&=-\xi^3-\eta^3,\end{aligned}$$
and thence
$$\begin{aligned}x_1^2x_2x_3z_2+\&\text{c.}&=9\xi^2\zeta^4+6\xi^4\zeta^2+\xi(\eta^3-\zeta^3)\cdot3\xi^2\zeta+6\xi\zeta\eta^3,\\ x_1^2z_2^2z_3^2+\&\text{c.}&=9\xi^4\zeta^4-\xi\zeta(\xi^3-\eta^3)=3\xi^2\zeta^4+6\xi^4\zeta^2/\ldots,\end{aligned}$$

and consequently

$$\Pi = \zeta^3(\eta^3 - \zeta^3) - \xi^3(-\xi^3 - \eta^3) - \zeta\xi(2\xi^2\zeta^2 - 2\xi\zeta^3 - 2\xi^3\zeta).$$

Now considering the equation

$$F \cdot U - \Pi = (\xi x + \eta y + \zeta z)^2(\xi' x + \eta' y + \zeta' z),$$

in order to find ξ', η', ζ' it will be sufficient to find the coefficients of x^3, y^3, z^3 in the function on the left-hand side of the equation. The coefficient of x^3 in Π is

$$\begin{aligned}&(x_1^2 + 2ly_1z_1)(x_2^2 + 2ly_2z_2)(x_3^2 + 2ly_3z_3)\\ &\quad = x_1^2x_2^2x_3^2 + 2l(x_1^2x_2^2y_3z_3 + \&\text{c.}) + 4l^2(x_1^2y_2z_2y_3z_3 + \&\text{c.}) + 8l^3y_1y_2y_3z_1z_2z_3;\end{aligned}$$

and it is easy to see that representing the function

$$(1, 1, 1, l \mid \beta\zeta - \gamma\eta,\ \gamma\xi - \alpha\zeta,\ \alpha\eta - \beta\xi)^3$$

by

$$(a, b, c, f, g, h, i, j, k, l \mid \alpha, \beta, \gamma)^3,$$

the symmetrical functions can be expressed in terms of the quantities $a, b, \&$c., and that the preceding value of the coefficient of x^3 in Π is

$$\begin{aligned}&a^2\\ &+ 2l(9hj - 6al)\\ &+ 4l^2(6gk - 3fj - 3hi + 3l^2)\\ &+ 8l^3\,bc;\end{aligned}$$

and substituting for $a, \&$c. their values, this becomes

$$\begin{aligned}&+ 2l\{-9(\zeta\eta^3 + 2l\eta\zeta^2)(\zeta^2\xi + 2l\xi\eta\zeta)\}\\ &- 4l^2\{-6(\zeta^2\xi + 2l\xi\eta\zeta)(\zeta\xi + 2l\xi\eta\zeta)\\ &\qquad + 3(\eta\zeta^2 + 2l\zeta\xi\eta)(\xi\eta + 2l\xi\eta\zeta)\\ &\qquad + 3(\zeta\eta^2 + 2l\eta\xi\zeta)(\eta\xi^2 + 2l\xi\eta\zeta)\},\end{aligned}$$

and reducing, we obtain for the coefficient of x^3 in Π the following expression,

$$\begin{aligned}&- 24l\xi^2\eta^3 + \xi^4 + \eta^4 \cdot \ldots\\ &- 24l^2(\eta^3\zeta^3 + \xi^3\zeta^3 + \xi^3\eta^3).\end{aligned}$$

Now the coefficient of x^3 in $F \cdot U$ is simply F, which is equal to

$$\begin{aligned}&\xi^4 + \eta^4 + \zeta^4 - 2\eta^2\zeta^2 - 2\zeta^2\xi^2 - 2\xi^2\eta^2\\ &\quad - 24l\,\xi\eta\zeta\ \ldots\\ &\quad - 24l^2(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta\\ &\quad - 32l^3(\eta^3\zeta^3 + \zeta^3\xi^3 + \xi^3\eta^3),\end{aligned}$$

and subtracting, the coefficient of x^3 in $F \cdot U - \Pi$ is

$$\begin{aligned}&\xi^4 - 2\eta^2\zeta^2 - 2\zeta^2\xi^2\\ &\quad - 6l\,\xi^2\eta\zeta\\ &\quad - 8l^3(\eta^3\zeta^3 + \zeta^3\xi^3 + \xi^3\eta^3)\\ &\quad - 8l^3(l^3 - \eta^3)(\zeta^3 - \xi^3)\\ &\quad - 48l^3\xi^3\eta^3\zeta^3,\end{aligned}$$

which is equal to

$$(1+8l^3)\,\xi^2(\xi^4-2\eta^3-2\zeta^3-6l\eta\zeta\xi^2).$$

The expression last written down is therefore the value of $\xi^2\xi'$, or dividing by ξ^2 we have ξ', and then of course the values of η', ζ' are known, and we obtain the identical equation

$$F\cdot U-\Pi=(1+8l^3)(\xi x+\eta y+\zeta z)^2\cdot\ldots,$$

and the second factor equated to zero is the equation of the satellite line of $\xi x+\eta y+\zeta z=0$.

33. The point of intersection of the line $\xi x+\eta y+\zeta z=0$ with the satellite line $\xi' x+\eta' y+\zeta' z=0$ is the satellite point of the former line; and the coordinates of the satellite point are at once found to be

$$x:y:z=(\eta^3-\zeta^3)(\eta\zeta+2l\xi^2)\,:\,(\zeta^3-\xi^3)(\zeta\xi+2l\eta^2)\,:\,(\xi^3-\eta^3)(\xi\eta+2l\zeta^2).$$

34. If the primary line $\xi x+\eta y+\zeta z=0$ is a tangent to the cubic, then (x_1,y_1,z_1) being the coordinates of the point of contact, we have

$$\xi:\eta:\zeta=x_1^2+2ly_1z_1\,:\,y_1^2+2lz_1x_1\,:\,z_1^2+2lx_1y_1;$$

these values give as before

$$\eta^3-\zeta^3=-(1+8l^3)x_1^3(y_1^3-z_1^3),$$

and they give also

$$\eta\zeta+2l\xi^2=(1+8l^3)y_1z_1,$$

and consequently we obtain

$$x:y:z=x_1(y_1^3-z_1^3)\,:\,y_1(z_1^3-x_1^3)\,:\,z_1(x_1^3-y_1^3),$$

that is, the satellite point of a tangent of the cubic is the point in which this tangent again meets the cubic.

Article Nos. 35 and 36. Theorems relating to the satellite point.

35. If the line $\xi x+\eta y+\zeta z=0$ be a tangent of the Pippian, then the locus of the satellite point is the Hessian.

Take (x,y,z) as the coordinates of the satellite point, then we have

$$x:y:z=(\eta^3-\zeta^3)(\eta\zeta+2l\xi^2)\,:\,(\zeta^3-\xi^3)(\zeta\xi+2l\eta^2)\,:\,(\xi^3-\eta^3)(\xi\eta+2l\zeta^2),$$

where the parameters ξ,η,ζ are connected by the equation

$$-l(\xi^3+\eta^3+\zeta^3)+(-1+4l^3)\,\xi\eta\zeta=0.$$

We have

$$x^3+y^3+z^3=\ldots,$$

and it is easy to see that the function on the right-hand side must divide by $\eta^3-\zeta^3$: hence $x^3+y^3+z^3$ will also divide by $\eta^3-\zeta^3$, and consequently by $(\eta^3-\zeta^3)(\zeta^3-\xi^3)(\xi^3-\eta^3)$.

We have

$$\begin{aligned}(\eta^3-\zeta^3)^3(\zeta^3-\xi^3)^3(\xi^3-\eta^3)^3&=-\xi^9-\eta^9-\zeta^9+3\xi^3(\eta^6+\zeta^6\cdot\ldots)-3\xi^6(\eta^3+\zeta^3)+\ldots\\&\quad+6\xi^3\eta^3\zeta^3\big(-\xi^3-3\eta^3-\eta^3+3l^2(\xi^3+\eta^3)-3\zeta\big)\\&\quad+12l^3\xi\eta\zeta\big\{-\xi\eta^2+\zeta^3+3\xi^3\eta^2-\eta\big\},\end{aligned}$$

and

$$\ldots$$

Adding these values and completing the reduction, we find

$$\begin{aligned}&(x^3+y^3+z^3)-(\eta^3-\zeta^3)(\zeta^3-\xi^3)(\xi^3-\eta^3)\\&\quad=-\xi^9-\eta^9-\zeta^9+l^9\eta^3\zeta^3+2\xi^9+2\xi^6\\&\qquad+18l\,\xi\eta\zeta\,\ldots\\&\qquad+12l^3(\xi^3+\eta^3+\zeta^3)\,\xi^3\eta^3\zeta^3,\end{aligned}$$

and we have also

$$\dots,$$

and thence

$$\begin{aligned}&\bigl\{A(x^3+y^3+z^3)+Bxyz\bigr\}\,\xi\eta\zeta\cdot(\eta^3-\zeta^3)(\zeta^3-\xi^3)(\xi^3-\eta^3)\\ &\quad=-A(\xi^9+\eta^9+\zeta^9)+\dots\\ &\quad+(12l^3A+8lB)(\xi^3+\eta^3+\zeta^3)\,\xi^3\eta^3\zeta^3\\ &\quad+(8lA+(1+8l^3)B)\,\xi^4\eta^4\zeta^4\\ &\quad+A\bigl\{(60l^3+81)\,\dots\bigr\}+l^4B(\eta^3\zeta^6+\zeta^3\xi^6+\xi^3\eta^6).\end{aligned}$$

The coefficient of $\eta^3\zeta^6+\zeta^3\xi^6+\xi^3\eta^6$ on the right-hand side will vanish if $A(1+2l^3)+l^4B=0$, or, what is the same thing, if $A=l^4$, $B=-(1+2l^3)$; and substituting these values, we obtain

$$\begin{aligned}&D\bigl\{l^2(x^3+y^3+z^3)-(1+2l^3)xyz\bigr\}\cdot(\eta^3-\zeta^3)(\zeta^3-\xi^3)(\xi^3-\eta^3)\\ &\quad=-l(\xi^3+\eta^3+\zeta^3)+(-l+4l^4)(\xi^3+\eta^3+\zeta^3)^2+(-l+8l^3-16l^6)\,\xi^3\eta^3\zeta^3,\end{aligned}$$

or, what is the same thing,

$$\begin{aligned}l^2(x^3+y^3+z^3)-(1+2l^3)xyz&=-(\eta^3-\zeta^3)(\zeta^3-\xi^3)(\xi^3-\eta^3)\\ &\times D\bigl\{-l(\xi^3+\eta^3+\zeta^3)+(-1+4l^3)\xi\eta\zeta\bigr\}^?.\end{aligned}$$

Hence the left-hand side vanishes in virtue of the relation between ξ,η,ζ, or we have

$$l^2(x^3+y^3+z^3)-(1+2l^3)xyz=0,$$

which proves the theorem.

36. Suppose that (X,Y,Z) are the coordinates of a point of the Hessian, and let (P,Q,R) be the coordinates of the point in which the tangent to the Hessian at the point (X,Y,Z) again meets the Hessian, or, what is the same thing, the satellite point in regard to the Hessian of the tangent at (X,Y,Z). And consider the conic

$$X(x^2+2lyz)+Y(y^2+2lzx)+Z(z^2+2lxy),$$

which is the first or conic polar of the point (X,Y,Z) in respect of the cubic. The polar (in respect to this conic) of the point (P,Q,R) will be

$$\xi x+\eta y+\zeta z=0,$$

where

$$\begin{aligned}\xi&=PX+l(RY+QZ),\\ \eta&=QY+l(PZ+RX),\\ \zeta&=RZ+l(QX+PY);\end{aligned}$$

or putting for (P,Q,R) their values,

$$\begin{aligned}\xi&=(Y^3-Z^3)(X^2-lYZ),\\ \eta&=(Z^3-X^3)(Y^2-lZX),\\ \zeta&=(X^3-Y^3)(Z^2-lXY);\end{aligned}$$

and if from these equations and the equation of the Hessian we eliminate (X,Y,Z), we shall obtain the equation in line coordinates of the curve which is the envelope of the line $\xi x+\eta y+\zeta z=0$. We find, in fact,

$$\xi^3+\eta^3+\zeta^3=(Y^3-Z^3)(Z^3-X^3)(X^3-Y^3)\times\Bigl\{l^2(X^3+Y^3+Z^3)-3l(X^3+Y^3+Z^3)XYZ+9l^2X^3Y^3Z^3+(1-4l^3)(Y^3Z^3+Z^3X^3+X^3Y^3)\Bigr\},$$

$$\xi\eta\zeta=(Y^3-Z^3)(Z^3-X^3)(X^3-Y^3)\times\Bigl\{l^3(X^3+Y^3+Z^3)XYZ+(1-l^3)X^3Y^3Z^3-l(Y^3Z^3+Z^3X^3+X^3Y^3)\Bigr\},$$

and thence recollecting that

$$HU = l^2(X^3+Y^3+Z^3) - (1+2l^3)XYZ,$$

we find

$$-l(\xi^3+\eta^3+\zeta^3) + (-1+4l^3)\,\xi\eta\zeta = -(Y^3-Z^3)(Z^3-X^3)(X^3-Y^3)\,(HU)^2,$$

and the equation of the envelope is

$$-l(\xi^3+\eta^3+\zeta^3) + (-1+4l^3)\,\xi\eta\zeta = 0,$$

which is therefore the Pippian. We have thus the theorem:

The envelope of the polar of the satellite point in respect to the Hessian of the tangent at any point of the Hessian, such polar being in respect of the conic which is the first or conic polar of the point of the Hessian in respect of the cubic, is the Pippian.

Article Nos. 37 to 40. Investigations and theorems relating to the first or conic polar of a point of the cubic.

37. The investigations next following depend on the identical equations

$$\begin{aligned}
&\big[\alpha(X^2+2lYZ)+\beta(Y^2+2lZX)+\gamma(Z^2+2lXY)\big]\\
&\qquad\times\big[-XYZ(x^3+y^3+z^3)+(X^3+Y^3+Z^3)xyz\big]\\
&=\big\{X(x^2+2lyz)+Y(y^2+2lzx)+Z(z^2+2lxy)\big\}\\
&\qquad\times\big[X(Y^3-Z^3)(\gamma y-\beta z)+Y(Z^3-X^3)(\alpha z-\gamma x)+Z(X^3-Y^3)(\beta x-\alpha y)\big]\\
&\quad+\big[x(X^2+2lYZ)+y(Y^2+2lZX)+z(Z^2+2lXY)\big]\\
&\qquad\times\big[-(\alpha YZ+\beta ZX+\gamma XY)(Xx^2+Yy^2+Zz^2)+(\alpha X^2+\beta Y^2+\gamma Z^2)(XYz+Yzx+Zxy)\big],
\end{aligned}$$

which is easily verified.

I represent the equation in question by

$$KT = WL + P\Theta;$$

then considering (x,y,z) as current coordinates, and (X,Y,Z) and (α,β,γ) as the coordinates of two given points Σ and Ω, we shall have $U=0$ the equation of the cubic, $W=0$ the equation of the first or conic polar of Σ with respect to the cubic, $P=0$ the equation of the second or line polar of Σ with respect to the cubic. The equation $T=0$ is that of a syzygetic cubic passing through the point Σ: the coordinates of the satellite point in respect to this syzygetic cubic of its tangent at Σ are

$$X(Y^3-Z^3) \,:\, Y(Z^3-X^3) \,:\, Z(X^3-Y^3);$$

and calling the point in question Σ', then $L=0$ is the equation of a line through the points Σ',Ω. The equation $\Theta=0$ is that of a conic, viz. the first or conic polar of Σ with respect to a certain syzygetic cubic

$$-2(\alpha YZ+\beta ZX+\gamma XY)(x^3+y^3+z^3)+(\alpha X^2+\beta Y^2+\gamma Z^2)xyz=0,$$

depending on the points Σ,Ω, or, what is the same thing, the conic $\Theta=0$ is a properly selected conic passing through the points of intersection of the first or conic polars of Σ with respect to any two syzygetic cubics; and lastly, K is a constant coefficient. The equation expresses that the points of intersection of

$$(W=0,\ P=0),\ (W=0,\ \Theta=0),\ (L=0,\ P=0),\ (L=0,\ \Theta=0),$$

lie in the syzygetic cubic $T=0$.

The left-hand side of the equation may be written

$$\begin{aligned}
&-XYZ\big\{\alpha(X^2+2lYZ)+\beta(Y^2+2lZX)+\gamma(Z^2+2lXY)\big\}(x^3+y^3+z^3+6lxyz)\\
&\quad+xyz\,\big\{\alpha(X^2+2lYZ)+\beta(Y^2+2lZX)+\gamma(Z^2+2lXY)\big\}(X^3+Y^3+Z^3+6lXYZ);
\end{aligned}$$

and it may be remarked also that we have

$$-3lXYZ\{\alpha(X^2+2lYZ)+\beta(Y^2+2lZX)+\gamma(Z^2+2lXY)\}$$

equal identically to

$$\begin{aligned}&l\big[X(Y^3-Z^3)(\gamma Y-\beta Z)+Y(Z^3-X^3)(\alpha Z-\gamma X)+Z(X^3-Y^3)(\beta X-\alpha Y)\big]\\&\quad-(\alpha YZ+\beta ZX+\gamma XY)(X^3+Y^3+Z^3+6lXYZ).\end{aligned}$$

Hence if we assume

$$X^3+Y^3+Z^3+6lXYZ=0,$$

the equation will take the form

$$KU=WL+P\Theta,$$

where the constant coefficient K may be expressed under the two different forms

$$\begin{aligned}K&=-XYZ\big[\alpha(X^2+2lYZ)+\beta(Y^2+2lZX)+\gamma(Z^2+2lXY)\big]\\&=\tfrac{1}{3}\big[X(Y^3-Z^3)(\gamma Y-\beta Z)+Y(Z^3-X^3)(\alpha Z-\gamma X)+Z(X^3-Y^3)(\beta X-\alpha Y)\big],\end{aligned}$$

and W, L, P, Θ have the same values as before. In the present case the point Σ is a point of the cubic: the equation $W=0$ represents the first or conic polar of the point in question, and the equation $P=0$ its second or line polar, which is also the tangent of the cubic. The line $L=0$ is a line joining the point Ω with the satellite point of the tangent at Σ, or dropping altogether the consideration of the point Ω, is an arbitrary line through the satellite point: the first or conic polar of Σ meets the cubic twice in the point Σ, and therefore also meets it in four other points; the conic $\Theta=0$ is a conic passing through these four points, and completely determined when the particular position of the line through the satellite point is given. And, as before remarked, $\Theta=0$ is a conic passing through the points of intersection of the first or conic polars of Σ with respect to any two syzygetic cubics. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at this point, and besides meets it in four other points; the four points in question are the points in which the first or conic polar of the given point in respect of the cubic is intersected by the first or conic polar of the same point in respect to any syzygetic cubic whatever.

38. The analytical result may be thus stated: putting

$$\kappa=\alpha YZ+\beta ZX+\gamma XY,\quad \lambda=\alpha X^2+\beta Y^2+\gamma Z^2,$$

or, if we please, considering κ, λ as arbitrary parameters, then the four points lie in the conic

$$\big(2\kappa X,\ 2\kappa Y,\ 2\kappa Z,\ -\lambda X,\ -\lambda Y,\ -\lambda Z \mid x,y,z\big)^2=0,$$

or, what is the same thing, they are the points of intersection of the two conics

$$\begin{aligned}\lambda x^2+Yy^2+Zz^2&=0,\\Xyz+Yzx+Zxy&=0.\end{aligned}$$

39. Considering the four points as the angles of a quadrangle, it may be shown that the three centres of the quadrangle lie on the cubic. To effect this, assume that the conic

$$\big(2\kappa X,\ 2\kappa Y,\ 2\kappa Z,\ -\lambda X,\ -\lambda Y,\ -\lambda Z \mid x,y,z\big)^2=0$$

represents a pair of lines; these lines will intersect in a point, which is one of the three centres in question. And taking x, y, z as the coordinates of this point, we have

$$x^2:y^2:z^2:yz:zx:xy=4\kappa^2YZ-\lambda^2X^2\ :\ 4\kappa^2ZX-\lambda^2Y^2\ :\ 4\kappa^2XY-\lambda^2Z^2\ :\ \lambda^2YZ+2\kappa\lambda X^2\ :\ \lambda^2ZX+2\kappa\lambda Y^2\ :\ \lambda^2XY+2\kappa\lambda Z^2;$$

and we may, if we please, use these equations to find the relation between κ, λ. Thus in the identical equation $x^2\cdot y^2-(xy)^2=0$, substituting for x^2, xy, y^2 their values, and throwing out the factor Z, we find

$(4\kappa^2-\lambda^2)\,XYZ-x^2y^2(X^3+Y^3+Z^3)=0$, and thence, in virtue of the equation $X^3+Y^3+Z^3+6lXYZ=0$, we obtain

$$4\kappa^2+6l\kappa\lambda^2-\lambda^3=0.$$

But the preceding system gives conversely,

$$X^2:Y^2:Z^2:YZ:ZX:XY=4\kappa^2yz-\lambda^2x^2\,:\,4\kappa^2zx-\lambda^2y^2\,:\,4\kappa^2xy-\lambda^2z^2\,:\,\lambda^2yz+2\kappa\lambda x^2\,:\,\lambda^2zx+2\kappa\lambda y^2\,:\,\lambda^2xy+2\kappa\lambda z^2.$$

Hence from the identical relation $X^2\cdot Y^2-(XY)^2=0$, substituting for X^2, XY, Y^2 their values, and throwing out the factor z, we find $(4\kappa^2-\lambda^2)xyz-\kappa\lambda^2(x^3+y^3+z^3)=0$, and thence, in virtue of the equation $4\kappa^2-\lambda^2=-6l\kappa\lambda^2$, we obtain

$$x^3+y^3+z^3+6lxyz=0,$$

which shows that the point in question lies on the cubic. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at the point, and meets it besides in four points, which are the angles of a quadrangle the centres of which lie on the cubic. In other words, the quadrangle is an inscribed quadrangle.

40. To find the equations of the three axes of the quadrangle, that is of the lines through two centres.

We have

$$\begin{aligned}
(4\kappa^2YZ-\lambda^2X^2)x+(\lambda^2XY+2\kappa\lambda Z^2)y+(\lambda^2ZX+2\kappa\lambda Y^2)z&=0,\\
(\lambda^2XY+2\kappa\lambda Z^2)x+(4\kappa^2ZX-\lambda^2Y^2)y+(\lambda^2YZ+2\kappa\lambda X^2)z&=0,\\
(\lambda^2ZX+2\kappa\lambda Y^2)x+(\lambda^2YZ+2\kappa\lambda X^2)y+(4\kappa^2XY-\lambda^2Z^2)z&=0;
\end{aligned}$$

or arranging these equations in the proper form and eliminating $\kappa^2, \kappa\lambda, \lambda^2$, we find

$$\begin{vmatrix} YZx, & Z^2y+Y^2z, & X(-Xx+Yy+Zz)\\ ZXy, & X^2z+Z^2x, & Y(Xx-Yy+Zz)\\ XYz, & Y^2x+X^2y, & Z(Xx+Yy-Zz)\end{vmatrix}=0;$$

or, multiplying out,

$$\begin{aligned}
&XYZ\{(Z^2-Y^2)x^3+(X^2-Z^2)y^3+(Y^2-X^2)z^3\}\\
&\quad+x^2yZY^2(-2X^2+Y^2+Z^2)+zx^2YZ^2(2X^2-Y^2-Z^2)\\
&\quad+y^2zXZ^2(-2Y^2+Z^2+X^2)+xy^2ZX^2(2Y^2-Z^2-X^2)\\
&\quad+z^2xYX^2(-2Z^2+X^2+Y^2)+yz^2XY^2(2Z^2-X^2-Y^2)=0.
\end{aligned}$$

We may simplify this result by means of the equation $X^3+Y^3+Z^3+6lXYZ=0$, so as to make the left-hand side divide out by XYZ: we thus obtain

$$\begin{aligned}
&(Z^3-Y^3)x^3+(X^3-Z^3)y^3+(Y^3-X^3)z^3\\
&\quad+(-3X^2Y-6lY^2Z)x^2y+(-3Y^2Z-6lZ^2X)y^2z+(-3Z^2X-6lX^2Y)z^2x\\
&\quad+(3XY^2+6lX^2Z)xy^2+(3YZ^2+6lY^2X)yz^2+(3ZX^2+6lZ^2Y)zx^2=0;
\end{aligned}$$

or in a different form,

$$\begin{aligned}
&(y^3-z^3)X^3+(z^3-x^3)Y^3+(x^3-y^3)Z^3\\
&\quad+(-3x^2y-6lz^2x)X^2Y+(-3y^2z-6lx^2y)Y^2Z+(-3z^2x-6ly^2z)Z^2X\\
&\quad+(3xy^2+6lyz^2)XY^2+(3yz^2+6lzx^2)YZ^2+(3zx^2+6lxy^2)ZX^2=0,
\end{aligned}$$

as the equation of the three axes of the quadrangle.

Article No. 41. Recapitulation of geometrical definitions of the Pippian.

In conclusion, I will recapitulate the different modes of generation or geometrical definitions of the Pippian, obtained in the course of the present memoir. The curve in question is:

1. The envelope of the line joining a pair of conjugate poles of the cubic (see Nos. 2 and 13).
2. The envelope of each line of the pair forming the first or conic polar with respect to the cubic of a conjugate pole of the cubic (see Nos. 2 and 14).
3. The envelope of a line which is the polar of a conjugate pole of the cubic, with respect to the conic which is the first or conic polar of the other conjugate pole in respect to any syzygetic cubic (see Nos. 2 and 9).
4. The locus of the harmonic with respect to a pair of conjugate poles of the cubic of the third point of intersection with the Hessian of the line joining the two conjugate poles (see Nos. 2 and 17).
5. The envelope of a line such that its lineo-polar envelope with respect to the cubic breaks up into a pair of lines (see No. 24).
6. The envelope of a line which meets three conics, the first or conic polars of any three points in respect to the cubic, in six points in involution (see No. 22).
7. The envelope of the second or line polar with respect to the cubic, of a point the locus of which is a certain curve of the sixth order in quadratic syzygy with the cubic and Hessian, viz. the curve $-S\cdot U^2+(HU)^2=0$ (see No. 27).
8. The envelope of a line having for its satellite point a point of the Hessian (see No. 35).
9. The envelope of the polar of the satellite point with respect to the Hessian of the tangent at a point of the Hessian, with respect to the first or conic polar of the point of the Hessian in respect to the cubic (see No. 36).

147.

A MEMOIR ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF AN EQUATION.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 489–499. Received December 18, 1856,—Read January 8, 1857.]

There are contained in a work, which is not, I think, so generally known as it deserves to be, the "Algebra" of Meyer Hirsch [the work referred to is entitled *Sammlung von Beispielen Formeln und Aufgaben aus der Buchstabenrechnung und Algebra*, 8vo. Berlin, 1804 (8 ed. 1853), English translation by Ross, 8vo. London, 1827] some very useful tables of the symmetric functions up to the tenth degree of the roots of an equation of any order. It seems desirable to join to these a set of tables, giving reciprocally the expressions of the powers and products of the coefficients in terms of the symmetric functions of the roots. The present memoir contains the two sets of tables, viz. the new tables distinguished by the letter (a), and the tables of Meyer Hirsch distinguished by the letter (b); the memoir contains also some remarks as to the mode of calculation of the new tables, and also as to a peculiar symmetry of the numbers in the tables of each set, a symmetry which, so far as I am aware, has not hitherto been observed, and the existence of which appears to constitute an important theorem in the subject. The theorem in question might, I think, be deduced from a very elegant formula of M. Borchardt (referred to in the sequel), which gives the generating function of any symmetric function of the roots, and contains potentially a method for the calculation of the Tables (b), but which, from the example I have given, would not appear to be a very convenient one for actual calculation.

Suppose in general

$$(1, b, c, \dots \mid 1, x)^n = (1-\alpha x)(1-\beta x)(1-\gamma x)\dots,$$

so that

$$-b=\Sigma\alpha,\quad +c=\Sigma\alpha\beta,\quad -d=\Sigma\alpha\beta\gamma,\quad \&c.,$$

and if in general

$$(pqr\dots)=\Sigma\,\alpha^p\beta^q\gamma^r\dots,$$

where as usual the summation extends only to the distinct terms, so that e.g. (p^2) contains only half as many terms as (pq), and so in all similar cases, then we have

$$-b=(1),\quad +c=(1^2),\quad -d=(1^3),\quad \&c.;$$

and the two problems which arise are, first to express any combination $b^p c^q \dots$ in terms of the symmetric functions $(l^\lambda m^\mu \dots)$, and secondly, or conversely, to express any symmetric function $(l^\lambda m^\mu \dots)$ in terms of the combinations $b^p c^q \dots$

It will conduce materially to brevity if $1^p 2^q \dots$ be termed the partition belonging to the combination $b^p c^q \dots$; and in like manner if $l^\lambda m^\mu \dots$ be termed the partition belonging to the symmetric function $(l^\lambda m^\mu \dots)$, and if the sum of the component numbers of the partition is termed the weight.

Consider now a line of combinations corresponding to a given weight, e.g. the weight 4, this will be

$$\begin{array}{cccccc} e & bd & c^2 & b^2c & b^4 & \text{(line)} \\ 4 & 13 & 2^2 & 1^2 2 & 1^4 & \end{array}$$

where I have written under each combination the partition which belongs to it, and in like manner a column of symmetric functions of the same weight, viz.

$$\begin{array}{ll} (4) & \text{(column)} \\ (31) & \\ (2^2) & \\ (21^2) & \\ (1^4), & \end{array}$$

where, as the partitions are obtained by simply omitting the ( ), I have not separately written down the partitions.

It is at once obvious that the different combinations of the line will be made up of numerical multiples of the symmetric functions of the column; and conversely, that the symmetric functions of the column will be made up of numerical multiples of the combinations of the line; but this requires a further examination. There are certain restrictions as to the symmetric functions which enter into the expression of the combination, and conversely, as to the combinations which enter into the expression of the symmetric function. The nature of the first restriction is most clearly seen by the following Table:

No. of Parts	Greatest Part	Comb.	Part.	Contain Mult. of the Sym. Functions	Greatest Part $\not>$	No. of Parts $\not<$
1	4	e	4	(1^4)	1	4
2	3	bd	13	(1^4), (21^2)	2	3
2	2	c^2	2^2	(1^4), (21^2), (2^2)	2	2
3	2	b^2c	$1^2 2$	(1^4), (21^2), (2^2), (31)	3	2
4	1	b^4	1^4	(1^4), (21^2), (2^2), (31), (4)	4	1

Thus, for instance, the combination bd (the partition whereof is 13) contains multiples of the two symmetric functions (1^4), (21^2) only. The number of parts in the partition 13 is 2, and the greatest part is 3. And in the partitions (1^4), (21^2) the greatest part is 2, and the number of parts is not less than 3. The reason is obvious: each term of the developed expression of bd must contain at least as many roots as are contained in each term of d, that is 3 roots, and since the coefficients are linear functions in respect to each root, the combination bd cannot contain a power higher than 2 of any root. The reasoning is immediately applied to any other case, and we obtain

First Restriction.—A combination $b^p c^q \dots$ contains only those symmetric functions $(l^\lambda m^\mu \dots)$, for which the greatest part does not exceed the number of parts in the partition $1^p 2^q \dots$, and the number of parts is not less than the greatest part in the same partition.

Consider a partition such as $1^2 2$, then replacing each number by a line of units thus,

$$\begin{array}{l} 1 \\ 1 \\ 1\,1 \end{array}$$

and summing the columns, we obtain a new partition 31, which may be called the conjugate[3] of $1^2 2$. It is easy to see that the expression for the combination $b^2 c$ (for which the partition is $1^2 2$) contains with

[3]The notion of Conjugate Partitions is, I believe, due to Professor Sylvester or Mr Ferrers. [It was due to Mr now Dr Ferrers.]

the coefficient unity, the symmetric function (31), the partition whereof is the conjugate of $1^2 2$. In fact $b^2 c = (-\Sigma\alpha)^2(\Sigma\alpha\beta)$, which obviously contains the term $1\alpha^3\beta$, and therefore the symmetric function with its coefficient $+1$ (31); and the reasoning is general, or

Theorem. A combination $b^p c^q \dots$ contains the symmetric function (partition conjugate to $1^p 2^q \dots$) with the coefficient unity, and sign $+$ or $-$ according as the weight is even or odd.

Imagine the partitions arranged as in the preceding column, viz. first the partition into one part, then the partitions into two parts, then the partitions into three parts, and so on; the partitions into the same number of parts being arranged according to the magnitude of the greatest part (the greatest magnitude first), and in case of equality according to the magnitudes of the next greatest part, and so on (for other examples, see the outside column of any one of the Tables). The order being thus completely defined, we may speak of a partition as being prior or posterior to another. We are now able to state a second restriction as follows.

Second Restriction.—The combination $b^p c^q \dots$ contains only those symmetric functions which are of the form (partition not prior to the conjugate of $1^p 2^q \dots$).

The terms excluded by the two restrictions are many of them the same, and it might at first sight appear as if the two restrictions were identical; but this is not so: for instance, for the combination bd^2, see Table VII (a), the term (41^2) is excluded by the first restriction, but not by the second; and on the other hand, the term $(3^2 1)$, which is not excluded by the first restriction, is excluded by the second restriction, as containing a partition $3^2 1$ prior in order to 32^2, which is the partition conjugate to 13^2, the partition of bd^2. It is easy to see why bd^2 does not contain the symmetric function $(3^2 1)$; in fact, a term of $(3^2 1)$ is $(\alpha^3\beta^3\gamma)$, which is obviously not a term of $bd^2 = (-\Sigma\alpha)(\Sigma\alpha\beta\gamma)^2$; but I have not investigated the general proof.

I proceed to explain the construction of the Tables (a). The outside column contains the symmetric functions arranged in the order before explained; the outside or top line contains the combinations of the same weight arranged as follows, viz. the partitions taken in order from right to left are respectively conjugate to the partitions in the outside column, taken in order from top to bottom; in other words, each square of the sinister diagonal corresponds to two partitions which are conjugate to each other. It is to be noticed that the combinations taken in order, from left to right, are not in the order in which they would be obtained by Arbogast's Method of Derivations from an operand a^n, a being ultimately replaced by unity. The squares above the sinister diagonal are empty (i.e. the coefficients are zero), the greater part of them in virtue of both restrictions, and the remainder in virtue of the second restriction; the empty squares below the sinister diagonal are empty in virtue of the second restriction; but the property was not assumed in the calculation.

The greater part of the numbers in the Tables (a) were calculated, those of each table from the numbers in the next preceding table by the following method, depending on the derivation of the expression for $b^{p+1}c^q \dots$ from the expression for $b^p c^q \dots$ Suppose, for example, the column cd of Table V (a) is known, and we wish to calculate the column bcd of Table VI (a). The process is as follows:

Given

$2^2 1$	21^3	1^5		
1	3	10		
321	2^3	$2^2 1^2$	21^4	1^6
1, 3, 2	3, 6, 12		10, 60	
1	3	3, 8, 22	60	

where the numbers in the last line are the numbers in the column bcd of Table VI (a). The partition $2^2 1$, considered as containing a part zero, gives, when the parts are successively increased by 1, the partitions 321, 2^3, $2^2 1^2$, in which the indices of the increased part (i.e. the original part plus unity) are 1, 3, 2; these numbers are taken as multipliers of the coefficient 1 of the partition $2^2 1$, and we thus have the new coefficients 1, 3, 2 of the partitions 321, 2^3, $2^2 1^2$. In like manner the coefficient 3 of the partition 21^3 gives the new coefficients 3, 6, 12 of the partitions 31^2, $2^2 1^2$, 21^4, and the coefficient 10 of the partition 1^5 gives the new coefficients 10, 60 of the partitions 21^4 and 1^6, and finally, the last line is obtained by addition. The process in fact amounts to the multiplication separately of each term of

$$cd = 1\,(2^2 1) + 3\,(21^3) + 10\,(1^5)$$

by $b = (1)$. It would perhaps have been proper to employ an analogous rule for the calculation of the combinations $cdef\ldots$ not containing b, but instead of doing so I availed myself of the existing Tables (b). But the comparison of the last line of each Table (a) (which as corresponding to a combination b^k was always calculated independently of the Tables (b)) with such last line as calculated from the corresponding Table (b), seems to afford a complete verification of both the Tables; and my process has in fact enabled me to detect several numerical errors in the Tables (b), as given in the English translation of the work above referred to. It is not desirable, as regards facility of calculation and independently of the want of verification, to calculate either set of Tables wholly from the other; the rules for the independent calculation of the Tables (b) are fully and clearly explained in the work referred to, and I have nothing to add upon this subject.

The relation of symmetry, alluded to in the introductory paragraph of the present memoir, exists in each Table of either set, and is as follows: viz. the number in the Table corresponding to any two partitions in the outside column and the outside line respectively, is equal to the number corresponding to the same two partitions in the outside line and the outside column respectively. Or, calling the two partitions P, Q, and writing for shortness, combination (P) for the combination represented by the partition P, and for greater clearness, symmetric function (P) (instead of merely (P)) to denote the symmetric function represented by the partition P, we have the following two theorems, viz.

Theorem. The coefficient in combination (P) of symmetric function (Q) is equal to the coefficient in combination (Q) of symmetric function (P);

and conversely,

Theorem. The coefficient in symmetric function (P) of combination (Q) is equal to the coefficient in symmetric function (Q) of combination (P).

M. Borchardt's formula, before referred to, is given in the 'Monatsbericht' of the Berlin Academy (March 5, 1855)[4], and may be thus stated; viz. considering the case of n roots, write

$$x^n \cdot \tfrac{1}{k}(1, b, c, \ldots \mid 1, x)^n = (1-\alpha x)(1-\beta x)\ldots(1-\kappa x) = fx,$$

then

$$\Sigma\Big(\frac{1}{1-\alpha x}\cdot\frac{1}{1-\beta y}\cdots\frac{1}{1-\kappa u}\Big) = \frac{1}{k^n}\cdot\frac{fxfy\ldots fu}{\Pi(x,y,\ldots u)}\cdot\frac{d}{dx}\frac{d}{dy}\cdots\frac{d}{du}\frac{\Pi(x,y,\ldots u)}{fxfy\ldots fu},$$

where $\Pi(x, y, \ldots u)$ denotes the product of the differences of the quantities $x, y, \ldots u$, and on the left-hand side the summation extends to all the different permutations of $\alpha, \beta, \ldots \kappa$, or what is the same thing, of $x, y, \ldots u$.

Suppose for a moment that there are only two roots, so that

$$(1, b, c \mid 1, x)^2 = (1-\alpha x)(1-\beta x),$$

then the left-hand side is

$$\frac{1}{(1-\alpha x)(1-\beta y)} + \frac{1}{(1-\alpha y)(1-\beta x)},$$

which is equal to

$$\begin{aligned}2 + (\alpha+\beta)(x+y) + (\alpha^2+\beta^2)(x^2+y^2) + 2\alpha\beta xy + (\alpha^3+\beta^3)(x^3+y^3)\\ + (\alpha^2\beta+\alpha\beta^2)(x^2y+xy^2) + \&\text{c.},\end{aligned}$$

and the right-hand side is

$$\frac{1}{c}\cdot\frac{fxfy}{x-y}\cdot\frac{d}{dx}\frac{d}{dy}\frac{x-y}{fxfy},$$

which is equal to

$$\frac{1}{c}\cdot\frac{1}{x-y}\Big\{\frac{f'xfy - f'yfx}{(fx)(fy)} + (x-y)\frac{f'xf'y}{fxfy}\Big\},$$

and therefore to

$$\frac{1}{c}\cdot\frac{1}{fxfy}\Big\{\frac{(f'x)fy - f'yfx}{x-y} + f'xf'y\Big\};$$

[4] And in Crelle, t. LIII, p. 195.—Note added 4th Dec. 1857, A. C.

or substituting for fx, fy their values,

$$\frac{f'xfy - f'yfx}{x-y}$$

becomes equal to

$$2c + bc(x+y) + 2c^2xy,$$

and $fxfy$ is equal to

$$b^2 + 2bc(x+y) + 4c^2xy.$$

The right-hand side is therefore equal to

$$\frac{2 + b(x+y) + 2cxy}{(1+bx+cx^2)(1+by+cy^2)},$$

and comparing with the value of the left-hand side, we see that this expression may be considered as the generating function of the symmetric functions of (α, β), viz. the expression in question is developable in a series of the symmetric functions of (x, y), the coefficients being of course functions of b and c, and these coefficients are (to given numerical factors *près*) the symmetric functions of the roots (α, β).

And in general it is easy to see that the left-hand side of M. Borchardt's formula is equal to

$$[0] + [1](1)(1)' + [2](2)(2)' + [1^2](1^2)(1^2)' + \&\text{c.},$$

where $(1), (2), (1^2), \&\text{c.}$ are the symmetric functions of the roots $(\alpha, \beta, \dots \kappa)$, $(1)', (2)', (1^2)', \&\text{c.}$ are the corresponding symmetric functions of $(x, y, \dots u)$, and $[0], [1], [2], [1^2], \&\text{c.}$ are mere numerical coefficients; viz. $[0]$ is equal to $1.2.3\dots n$, and $[1], [2], [1^2], \&\text{c.}$ are such that the product of one of these factors into the number of terms in the corresponding symmetric function $(1), (2), (1^2), \&\text{c.}$ may be equal to $1.2.3\dots n$. The right-hand side of M. Borchardt's formula is therefore, as in the particular case, the generating function of the symmetric functions of the roots $(\alpha, \beta, \dots \kappa)$, and if a convenient expression of such right-hand side could be obtained, we might by means of it express all the symmetric functions of the roots in terms of the coefficients.

Tables relating to the Symmetric Functions of the Roots of an Equation.

The outside line of letters contains the combinations (powers and products) of the coefficients, the coefficients being all with the positive sign, and the coefficient of the highest power being unity; thus in the case of a cubic equation the equation is

$$x^3 + bx^2 + cx + d = (x-\alpha)(x-\beta)(x-\gamma) = 0.$$

The outside line of numbers is obtained from that of letters merely by writing $1, 2, 3\dots$ for $b, c, d\dots$, and may be considered simply as a different notation for the combinations. The outside column contains the different symmetric functions in the notation above explained, viz. (1) denotes $\Sigma\alpha$, (2) denotes $\Sigma\alpha^2$, (1^2) denotes $\Sigma\alpha\beta$, and so on. The Tables (a) are to be read according to the columns; thus Table II (a) means $b^2 = 1\,(2) + 2\,(1^2)$, $c = (1^2)$. The Tables (b) are to be read according to the lines; thus Table II (b) means $(2) = -2c + 1b^2$, $(1^2) = +1c$.

I (a). I (a) and I (b) (weight 1)

	1 b
(1)	−1

	1 b
(1)	−1

II (a) and II (b) (weight 2)

	2	1^2
	c	b^2
(2)	+1	+2
(1^2)		+1

	2	1^2
	c	b^2
(2)	−2	+1
(1^2)	+1	

III (a) and III (b) (weight 3)

	3	12	1^3
	d	bc	b^3
(3)	−1	−3	−1
(21)		−1	−3
(1^3)			−1

	3	12	1^3
	d	bc	b^3
(3)	−3	+3	−1
(21)	+3	−1	
(1^3)	−1		

[Tables IV (a)/(b) (weight 4) through X (a)/(b) (weight 10) follow in the printed memoir. They are large rectangular grids whose rows are indexed by the partitions of the weight and whose columns are indexed by the conjugate combinations of coefficients $b, c, d, e, f, g, h, i, k$. Each entry is an integer (with sign) giving the coefficient that links a power-product of coefficients to a symmetric function $(l^\lambda m^\mu \ldots)$ as explained above; many squares above the sinister diagonal are zero (empty). The tables span PDF pages 424–434 in this reprint (journal pages 408–418), beginning with the simple weight-4 layout shown above and growing to weight 10 (Table X), in which the column heading contains terms like bch, b^2h, eg, b^2g, bdg, b^4g, etc., and the partition column ranges from (10) through (1^{10}).]

[Table IV (a) (weight 4)]. Column headings, left to right: e | bd | c^2 | b^2c | b^4. Row entries (partition rows (4), (31), (2^2), (21^2), (1^4)):

	e	bd	c^2	b^2c	b^4
(4)	+1	+4	+2	+6	+24
(31)		+1	+2	+5	+12
(2^2)			+1	+2	+6
(21^2)				+1	+4
(1^4)					+1

Table IV (b) (weight 4) inverts this: each row gives a symmetric function as a signed combination of the coefficient-products e, bd, c^2, b^2c, b^4:

$$\begin{aligned}
(4) &= -4e + 4bd + 2c^2 - 4b^2c + 1b^4,\\
(31) &= +4e - 1bd - 2c^2 + 1b^2c,\\
(2^2) &= +2e + 1c^2,\\
(21^2) &= -4e + 1bd,\\
(1^4) &= +1e.
\end{aligned}$$

[Tables V (a)/(b) (weight 5), VI (a)/(b) (weight 6), VII (a)/(b) (weight 7), VIII (a)/(b) (weight 8), IX (a)/(b) (weight 9), and X (a)/(b) (weight 10) extend the same pattern with progressively wider grids; the printed memoir reproduces them in full. They are large numerical tables, occupying PDF pages 426–434 (journal pages 410–418), with row-headings the partitions of the weight in the conventional Cayley order (greatest part first, ties broken by next greatest part, etc.) and column-headings the conjugate coefficient-products $f, be, cd, b^2d, bc^2, \ldots$ for weight 5; $g, bf, ce, b^2e, c^2, bcd, b^3d, c^3, b^2c^2, b^4c, b^6$ for weight 6; and so on through weight 10. The entries are signed integers; the symmetry theorem stated above asserts that the entry corresponding to the row-partition P and column-partition Q in Table N (a) equals the entry in row Q, column P of the same table.]

The numerical content of Tables IV–X (running ∼300 entries per table by Table X) is preserved here in editorial summary because the original layouts are wide multi-column grids that resist reflow at 11 pt without compromising line-by-line readability; the original journal version (Phil. Trans. 1857, Tables IV (a)–X (b)) remains the authoritative source for the precise numerical values.